

STRATA: A Stratified Process-Mining Framework for HVAC Fault Detection and Diagnosis

Yassh Singh

Supervisors: Dr. Anthony Aighobahi, Dr. Mridula Sharma
Thompson Rivers University — UREAP 2026

September 2026

Abstract

We report STRATA, a stratified process-mining framework for building HVAC fault detection and diagnosis, together with the falsification protocol under which it was built: hypotheses pre-registered with the observation that would refute them, results recomputed independently from raw data before they are quoted, and automated tests that pin published numbers, fired falsifiers, and disclosure obligations. The framework abstracts ordinary one-minute trend data into an event log along the control sequence, discovers process models at device and unit strata, and scores each day through eight calibrated, physically interpretable channels against a measured noise floor. On five public LBNL benchmark systems — two never previously benchmarked — it detects 158 of 175 seeded faults, 157 after one dataset defect is adjudicated (14 of 14 with 13 adjudicated, 26 of 30, 25 of 29, 50 of 55 and 43 of 47), at a median of one day and at 26 false-alarm days in 480 held-out fault-free days (1.0–9.4% per system; 15.6% on the air handler with every channel counted; every rate out-of-sample), the fourth and fifth systems onboarded blind by configuration alone and the last sixteen detections recovered by a pre-registered audit that mapped every recorded column. Each alarm carries a location: the correct terminal unit is named in 37 of the 50 detections with determinable ground truth on the two fan-powered systems, none in 13, and a wrong one in none, and the faulted subsystem in 83% of all 158 detections. The false-alarm budget holds under injected sensor error, change-of-value logging and schedule shifts on all five systems (worst 7 of 96 days) and on one measured rooftop unit (3 of 49). Process mining helps to get detection but is not detecting on its own: the strata and the event abstraction carry the localisation, invariants that discovery surfaced became several of the rules that detect, whereas conformance checking of the discovered nets, pre-registered as the detector, was refuted — a one-line hand-written rule out-covered it, and an ablation over 199 scenarios on six systems finds no fault detected only by the discovered models (a one-sided 95% upper bound of 1.5% on the sole-detection rate), so removing them changes no scorecard and lowers three systems’ false-alarm rates. A sealed test of whether the models predict where a rule transfers between buildings never had attainable power, as its third amendment records, and is reported as a protocol failure, not a result. A principal-component baseline detects three fewer of the 73 scenarios it shares, inside overlapping intervals, with fewer false alarms on the two terminal-unit systems, and cannot name the device. A third-party Guideline 36 rule battery, held to the same gate, is within one detection and one false-alarm day of STRATA on the air handler and detects about a third as many faults on the terminal units at no false alarms; the pre-registered improvement claim over current practice therefore fired its falsifier and is withdrawn on the air handler. The evaluation also produced eight previously unreported, machine-verified errata in the benchmark datasets. Every number regenerates from the public repository.

1 Introduction

Buildings account for approximately 34% of global energy-related carbon dioxide emissions, and HVAC systems carry the dominant share within that total [5]. The fraction of HVAC energy wasted by undetected faults has consistently been estimated at 15–30% across recent reviews [5, 40], but until late 2023 these numbers rested on relatively small studies. The first large-scale empirical characterisation appeared when Crowe and colleagues analysed multi-year fault-detection output across more than 60,000 pieces of HVAC equipment in 317 commercial buildings [10]. Their study reports an average of 245 faults per building per month and finds 40% of all air handling units in a faulted state on any given day. The most common AHU fault in their dataset, a supply-air-temperature setpoint that is never reset, was observed on 55% of the AHUs surveyed. These buildings were not unmonitored; many run modern building automation systems with alarms and fault-detection features, yet the faults persisted, which points to a gap in the kind of fault these tools can see rather than in their deployment.

The nature of that gap is specific. Of the ten most common AHU faults catalogued by Crowe and colleagues, seven are behaviour-based rather than condition-based: incorrect schedules, unreleased manual overrides, missed setpoint resets, and broken economiser sequences manifest as deviations in the temporal ordering of system states rather than as isolated component failures. A threshold alarm fires when a single variable leaves its acceptable band, and a pre-programmed rule fires when a named fault pattern is matched, but neither mechanism detects a control sequence that executes its steps in the wrong order or skips a step entirely unless a rule was written for that specific sequence in advance — which is possible, and has been done for logic faults [21], but only for the sequences someone anticipated. The structural argument that follows is straightforward: detection methods that model the sequence in which a building moves between operating states have a natural advantage on the fault classes that dominate the empirical landscape.

The dominant research methodology for HVAC fault detection does not model that sequence either. Machine learning has been surveyed across 211 AI-based HVAC FDD studies published between 2013 and 2023 [5], with reported accuracies on the standard benchmarks saturating at 97–99% across hybrid random-forest pipelines on RP-1312 data [44] and convolutional-and-LSTM hybrids on real-building data [39]. The most recent work has shifted toward interpretability, but in a specific way: a deep model is paired with a post-hoc explanation method such as layer-wise relevance propagation or attention attribution [12, 52]. The resulting explanation is at the level of input features rather than at the level of the control sequences the building was supposed to follow. A facility manager told that sensor *X* had high relevance for a prediction knows that something is unusual about that signal, but is not told how the building was supposed to be behaving instead.

A second problem concerns reproducibility. A recent audit of 65 primary machine-learning HVAC fault detection studies, circulated as a preprint, found two that shared code links, one of them broken, and that 72% do not specify whether their dataset is public, proprietary, or commercial [32]. The field has plateaued on accuracy, a metric that does not measure what an operator needs, with little of the existing work independently verifiable, and with the dominant explanation style answering a question (which signal mattered) different from the one operators ask (what should the system have been doing).

The opportunity is to change the model class. Process mining is a methodology in which a discrete event log derived from system observations is used to discover a Petri-net model of expected behaviour, with conformance checking against that model producing an alignment-based deviation signal that carries formal fitness and precision semantics [45]. We apply this approach to HVAC fault detection in stratified form, treating sensor-derived events as the input log and ASHRAE Guideline 36 control sequences as the conceptual reference: events are abstracted from the control

sequence at device and unit granularity, models are discovered per stratum, and the channels that read the resulting log — absence, frequency and conformance — sit beside physics rules, residuals and an oscillation channel that read the raw signals, each calibrated on its own noise. Our initial hypothesis was that the discovered Petri net would serve as both the expected-behaviour model and the detector. That hypothesis was pre-registered, tested, and refuted (Section 7): a single hand-written rule out-covered the unit-stratum model channel on the scenario where that channel was strongest. What the refutation settled is the division of labour. The stratified structure carries the detections: the event abstraction and the strata give every alarm its location, the absence and frequency channels read the log directly, and several of the rules that detect were written from invariants that discovery surfaced; conformance checking of the discovered nets contributes no detection of its own. Process mining helps to get detection but is not detecting on its own. Whether the models can predict, in advance, where such a rule will hold on another building is a further claim this paper set out to test and could not (Section 7). Detection is done by calibrated, physically interpretable channels against a measured noise floor.

Four contributions follow. First, the framework and the detector it yields: eight calibrated channels over a stratified event log that report at a stated false-alarm budget rather than at accuracy alone, evaluated on five public benchmark systems, two of which had not previously been benchmarked, onboarded by configuration alone, naming the implicated unit, with one alarm shown verbatim. Second, a settled answer to the process-mining question that motivated the work: one pre-registered negative result — discovered models detect no fault that calibrated channels miss, so their contribution is the event log, the strata and the invariants they surface rather than detection of their own — and one pre-registered test of whether they predict where a hand-written rule transfers, which never had attainable power, as its third amendment records, and which is reported as a protocol failure rather than as a result in either direction. A precedent for the shape of this result exists in another cyber-physical domain: Zhong and Lisitsa report, in a preprint, that naïve process mining on network event logs was ineffective for intrusion detection [56]. This remains, to our knowledge, the first application of event-log process mining to building HVAC fault detection, corroborated by the one systematic review of process mining on sensor data whose scope could have captured such work [6]; the claim is that the application was made and tested, and that what it contributes — structure and localisation, not detection by conformance alone — is stated. Third, the evaluation protocol — pre-registration, honoured falsifiers, adversarial recomputation, tests that pin disclosures, and full artefact regeneration — in a subfield where an audit of sixty-five primary studies found two that shared code links, one of them broken [32]. Fourth, eight machine-verified errata in the benchmark datasets. Deployment on an operating building remains the intended next phase.

2 Research Aim and Objectives

The question this research set out to answer is whether process models discovered from fault-free operation can detect faults in building HVAC systems interpretably and portably, and what survives when they cannot. The work reported here is validated on public benchmark data; deployment on an operating building is the intended next phase rather than a result of this paper.

What the framework is meant to do. A building operator today watches trend graphs and waits for a complaint. STRATA is meant to consume the building’s ordinary one-minute trend data and one configuration file that names its sensors and control rules, learn from a fault-free period what healthy operation looks like, and then, every day, name the days that break a physical relationship

the healthy period established: which unit, which relationship, by how much, and how often that channel has cried wolf on healthy days. The output is a work order, not a graph, with a false-alarm budget the operator can hold the system to. Onboarding a new equipment type should be a configuration exercise, not a software project. That intent fixes the six objectives below, each of which is reported with what was actually done and where it now stands.

1. Abstract events from sensor data. Objective: convert continuous HVAC sensor streams into discrete event logs using control-sequence rules grounded in ASHRAE Guideline 36, with fault-signature events held out of model discovery so that the models cannot re-detect the rules that produced them. Done: an event-abstraction layer driven entirely by configuration, applied unchanged to seven LBNL datasets (five simulated systems, a simulated and a measured rooftop unit); the alphabet wall between state and fault-signature events is enforced by a build-time invariant test, and the circularity it prevents was measured rather than asserted (Section 4). Status: met.
2. Discover models and locate invariants. Objective: learn Petri-net models of expected control behaviour from fault-free data at device, unit and cross-system granularity, and establish what those models contribute. Done: nets were discovered at all three strata on every system, and their contribution was measured under pre-registration: conformance against them is credited with no detection that the calibrated channels miss, on 199 scenarios across six systems, with a positive control showing the channel can see order on two systems, a seven-instrument grid showing the null is not the miner's, and every recorded column finally read. Their contribution was at design time: invariants they surfaced became rules that detect (Section 7). Status: met, in the form its falsification required — the models' contribution is established and stated (structure, event log and invariants at design time; no detection by conformance of their own), and the founding hypothesis is refuted.
3. Detect against a measured noise floor. Objective: calibrate interpretable channels on fault-free data alone and require every alarm to exceed that channel's own measured noise, so that detection is reported with a stated false-positive budget rather than with accuracy alone. Done: eight channels, every threshold from the fault-free year, a three-way calendar split so that every false-alarm rate is out-of-sample, and a per-day noise floor per channel; the budget held under injected field conditions and on one measured building (3 of 49 held-out days). Status: met; the cost is stated where it bit (the statistical repair removed seven detections, Section 9).
4. Benchmark against known answers. Objective: evaluate on the LBNL benchmark family against documented ground-truth faults, and compare with a machine-learning baseline on identical scenario universes and with a third-party implementation of the ASHRAE Guideline 36 rule battery. Done: 158 of 175 seeded scenarios detected on the five simulated systems (14 of 14 with 13 adjudicated, 26 of 30, 25 of 29, 50 of 55, 43 of 47; 157 under the adjudication) at 26 false-alarm days in 480, most on the first day; against an audited principal-component baseline on identical universes, 64 of 73 to its 61, with three scenarios only STRATA catches and none only the baseline catches; the seventeen remaining misses are coil fouling whose recorded values sit inside every healthy band, and one is visible but uncertifiable from a single healthy year (Section 6.4). Done, against the claim: the pre-registered comparison with the open-fdd implementation of the Guideline 36 battery (Section 9.1) fired its own falsifier: held to the treatment STRATA gives its own signature rules, the battery detects 12 of 14 on the single-duct air handler at one false-alarm day against STRATA's 13 at one, and

8 of 30 and 9 of 29 on the terminal units at no false-alarm day against STRATA’s 26 and 25 at eight and nine. Status: all three parts executed; the improvement claim over current practice is withdrawn on the air handler and stands on the terminal units only as a coverage difference bought with false alarms.

5. Make every claim checkable. Objective: pre-register each hypothesis with the condition that would refute it, subject every result to adversarial recomputation before it is quoted, and guard published values, fired falsifiers and disclosure obligations with automated tests. Done: more than thirty pre-registered experiments (Appendix A), ten adverse results honoured and published together with four later fired falsifiers reported beside the results they bound, every headline number regenerated from a committed artefact under a regression gate, and a test suite that fails the build if a published caveat is dropped. The protocol caught our own errors: a drift between the library and the scripts, a statistical floor that was looser than the one it replaced, and sixteen “unreachable” misses that were columns nobody had mapped. Status: met; this is the contribution the paper is surest of.
6. Release the work openly. Objective: publish the pipeline and its building configurations as open-source software, evaluated on public data with a documented path from a fresh clone to regenerated artefacts. Done: the repository, every configuration, every artefact and the reproduction path are public; eight machine-verified defects in the benchmark datasets are published with reproducing commands (Section 8). Status: met.

What the objectives add up to: a detector that does what the intent above asks on the benchmarks it was tested on, a settled and well-defended answer to the process-mining question that motivated it, and a protocol that made both claims checkable. What they do not yet add up to is a field result: the measured building contributes a false-alarm rate, and its single fault case was not visible in its recorded signals.

Two goals lie outside this scope and are stated as future work rather than quietly dropped. The first is a hybrid extension combining process-mining features with machine-learning classifiers, deferred until the negative result about the models’ own detection had been established. The second is deployment on an operating building, which needs a healthy period to learn from and the sensors mapped; the coverage audit of Section 6.4 is the checklist that deployment would start with.

3 Related Work

3.1 Machine learning for HVAC fault detection, and the move from accuracy to interpretability

The machine-learning era of HVAC fault detection is well documented. Bi and colleagues survey 211 AI-based HVAC FDD studies published between 2013 and 2023, organising them into traditional machine-learning, deep-learning, hybrid AI, and physical-model branches [5]. Reported accuracies on the standard benchmark datasets have effectively saturated. Tun and colleagues reached 98% accuracy across thirteen fault types using a hybrid random-forest and support-vector-machine pipeline on RP-1312 AHU data [44]. Prince, Yoon, and Kumar reached 97% accuracy on data from three real buildings using a thirteen-layer one-dimensional convolutional-and-LSTM hybrid that remains robust to Gaussian noise and to 10% missing values [39]. The authors of that paper close by explicitly identifying interpretability as the problem they did not solve, proposing grafting SHAP or LIME as future work.

The most directly relevant interpretable benchmark for the LBNL SDAHU subset is Deng and colleagues’ adaptive fusion graph convolutional network. The model reaches 88.09% accuracy at 500 labelled samples and 75.15% at only seven samples, with per-node sensor importance as the explanation mechanism [12]. Subsequent work in the same vein pairs multi-head attention with parallel dilated convolutions and applies layer-wise relevance propagation for explanation on chiller fault data, reaching an F1 score of 0.9841 across 27 operating conditions [52]. The pattern across this body of work is consistent: a deep black-box network is the model, a post-hoc analysis layer provides the explanation, and the explanation is framed at the level of input features rather than control flow. The architectural opportunity is for a method whose model is its explanation.

3.2 Unsupervised methods and cross-building deployment

Real buildings rarely come with ground-truth fault labels, and the strongest recent work on unlabelled data uses self-supervised or autoencoder methods. Abdollah and colleagues pretrained a Transformer encoder via masked-token reconstruction on unlabelled HVAC time-series from a heat-pump building at the Polytechnic University of Milan, then used Peak-Over-Threshold detection on reconstruction error to flag a real two-month monitoring fault between October and December [1]. The most directly relevant precedent for a Canadian-university real-BAS deployment is El Mokhtari and McArthur’s autoencoder-based work at Toronto Metropolitan University, which distinguishes equipment-level from system-level faults across five fan coil units and demonstrates cross-unit transfer learning [13]. These methods share a common limitation. Each produces a learned latent embedding whose deviation signal is a scalar anomaly score, not a process-level explanation.

The cross-building generalisation problem is the binding constraint that prevents most existing HVAC FDD work from being deployed in the field. The established response is domain adaptation, which transfers learned network weights: Zhu and colleagues migrate fault detection between two different building chillers [57], and Ghalamsiah and colleagues apply a contrastive adaptation network to ASHRAE-1312 simulation as a labelled source and real AHU measurements as an unlabelled target [15]. What STRATA transfers between buildings is a configuration file naming sensors and rules, not a set of learned weights; transferring it is therefore a different operation from the weight transfer the literature above studies, and Section 7 reports that the discovered models themselves did not transfer.

3.3 Process mining: foundations, adjacent successes, and the empty quadrant in building energy

Process mining was formalised by van der Aalst as a methodology for discovering process models from event logs, checking observed behaviour for conformance, and enhancing the resulting models with additional perspectives [45]. Conformance checking was used directly as a trace-level anomaly detector on event logs [4, 41]; the line subsequently moved to neural sequence models that outperform conformance-based baselines on standard log corpora [34]. The conformance channels used here are the nearest descendant of the former, and the ablation reported in Section 6 — no fault detected only by them — is consistent with the motivation for the latter. The methodology has also been applied across cyber-physical and sensor-driven domains adjacent to building energy. Vitale and colleagues applied a full process-mining pipeline to fault diagnosis on the RoAD robotic-arm dataset, reaching an F1 score of up to 98.925% with a conformance-checking time as low as 0.020 seconds [50]. The same group then demonstrated that combining process-mining-derived alignment features with downstream dimensionality reduction substantially outperforms pure con-

formance checking across four cross-domain datasets, with F1 improvements ranging from 30 to 70 points [51]. Husom and colleagues’ UDAVA pipeline extended this hybrid pattern to industrial Internet-of-Things sensor validation [23]. Methodologically the closest analogue to building HVAC is Moradbeikie and colleagues’ work on pasteurisation, where continuous multivariate sensor signals are abstracted into discrete activity sequences via hidden Markov models and conformance is checked against HACCP critical control points [30]. Pasteurisation, like HVAC operation, is a continuous-flow process governed by an explicit regulatory standard; the analogy between HACCP critical control points and ASHRAE Guideline 36 control sequences is suggestive. It is not a direct port: pasteurisation has a single physical flow and a case notion given by the batch, whereas HVAC operation has no natural case boundary, concurrent devices at several granularities, and an alphabet in which some events already encode fault suspicion — both the case notion used here (the calendar day) and the partition of the event alphabet had to be constructed. The technical barrier common to all these applications is event abstraction, the conversion of continuous sensor time-series into discrete activity sequences, which was formalised in the methodology paper by Janssen and colleagues [24].

Despite these adjacent successes, building HVAC remains an empty quadrant in the literature. The novelty claim rests principally on one review and is corroborated by three others whose scope or period would not necessarily have captured such work. Bi and colleagues’ survey of 211 AI-based HVAC FDD studies contains no process-mining entries in any leaf of its method taxonomy [5]. Akhramovich and colleagues’ review of 47 process-mining studies in Industry 4.0 covers manufacturing, internal logistics, mining, alarm management, supply-chain logistics, and product lifecycle management; building applications do not appear [2]. Where process mining has reached the built environment at all it is for construction and facility-management workflows [29], with cases as projects and activities as human tasks; the operational control flow of a building’s mechanical systems is not addressed. Brzychczy and colleagues’ more recent review of 36 process-mining studies on sensor data documents the same absence at finer granularity: fourteen industrial applications, eleven smart-home applications (all motion-sensor occupant-activity recognition), seven healthcare applications, two smart-office applications, two smart-product applications, and one other, with no building HVAC [6]. The Lawrence Berkeley National Laboratory team’s canonical FDD-methods taxonomy from 2017 organises automated fault-detection methods into quantitative model-based, qualitative model-based, and process-history-based categories, and process mining is not present at any leaf [20]. The absence is consistent across all four, and our own exact-phrase searches of the OpenAlex corpus for process mining or conformance checking in conjunction with HVAC, building automation, air handling, or chillers return no work; nor did the ICPM 2025 programme, the most recent at the time of writing, contain any. No published work, to our knowledge, applies event-log process mining to building HVAC fault detection.

3.4 Rule-based fault detection in practice, and what process mining adds

Fault detection is not absent from real buildings; it is simply rule-based. The most widely deployed form is the fault-detection capability embedded in commercial building automation systems. Modern platforms ship libraries of pre-programmed fault rules: the Automated Logic WebCTRL system at the building intended for this work’s deployment phase, for example, includes a standard library that matches over one hundred named faults in common HVAC equipment.¹ The characterisation survey of fourteen commercial FDD tools by Granderson and colleagues confirms that this rule-based and threshold-based style is the norm across the commercial landscape [20], and Guideline

¹Automated Logic, Fault Detection and Diagnostics, <https://www.automatedlogic.com/en/products/webctrl-building-automation-system/integration/fdd-reporting-and-dashboards/>.

36’s fault conditions have themselves been used as the reference against which learned diagnostic models are built [38]. The strength of a rule library is that it works immediately on connection and requires no baseline data; its structural limitation is that it can only detect the fault patterns that were coded into it in advance, and so it misses novel failure modes and the gradual, multi-variable, sequence-level deviations that no rule author anticipated. Section 9.1 measures both the strength and the limitation directly, against a third-party implementation of the Guideline 36 battery on the same benchmarks.

The same rule-based philosophy characterises the most directly comparable research programme, the Lawrence Berkeley National Laboratory team’s automated fault-correction line. Pritoni and colleagues field-tested seven hand-coded correction algorithms across 225 distinct equipment items in four buildings on three building-automation-system platforms, targeting common control problems such as incorrectly programmed schedules, unreleased manual overrides, and missed setpoint resets [40]. Each algorithm is a pre-encoded rule matched against BAS data and applied through a two-way BACnet writeback to the building’s control system. Both the commercial libraries and the LBNL correction algorithms share the assumption that the faults of interest can be enumerated and encoded ahead of time.

STRATA makes the opposite assumption and is therefore complementary to both. Rather than matching observed data against a fixed catalogue of known faults, it learns from the building’s own fault-free operating data where that building’s behaviour is invariant, and calibrates detection channels to hold those invariants. Conformance alignment against the discovered models is one such channel rather than the whole detector — a distinction our own results forced on us (Section 7) — and it contributes the property a rule library cannot supply unaided: during development the invariants were surfaced by discovery rather than enumerated in advance, which is how several of the rules now deployed came to be written. Every channel carries a measured false-alarm budget, which is what makes the resulting alarms answerable rather than merely numerous. A practical consequence is that STRATA can run on the same trend data a commercial system such as WebCTRL already records, addressing faults that fall outside the rule library or below its thresholds — on the benchmarks used here, the terminal-unit faults that Section 9.1 measures; the behaviour-based faults the introduction cites are not seeded in any of them (Section 9) — while the rule library continues to handle the well-understood patterns it was built for. In principle the located, evidenced alarms STRATA produces could serve as input to the LBNL correction algorithms, extending automated correction beyond the patterns pre-encoded by hand. The LBNL FDD dataset family [19] provides the labelled simulation testbed that supports this complementary positioning.

One adjacent body of work must be distinguished as a class, because its vocabulary is similar. Hand-designed Petri nets have modelled HVAC and building systems for two decades — for hybrid system modelling [49], for building-automation monitoring [17], and for chiller fault diagnosis [54] — and in every case the net is authored by an engineer rather than discovered from a log. The most recent example is control: Deabes, Bouazza, and Alghami designed a fuzzy Petri-net controller for HVAC temperature regulation, hand-engineering both the Petri-net structure and the fuzzy rules and using the model as a forward control device for an air-conditioning compressor [11]. None of this work contains an event log, a process-discovery algorithm, or alignment-based conformance checking; the diagnosis variant matches observations against a net an expert drew. The present work does the methodological opposite: a Petri net is discovered from a real event log of HVAC operation, then used as a reference against which alignment-based conformance checking diagnoses operational deviations. Hand-designed Petri-net controllers and data-discovered Petri-net diagnostic models are two distinct uses of the Petri-net formalism, and they should not be conflated.

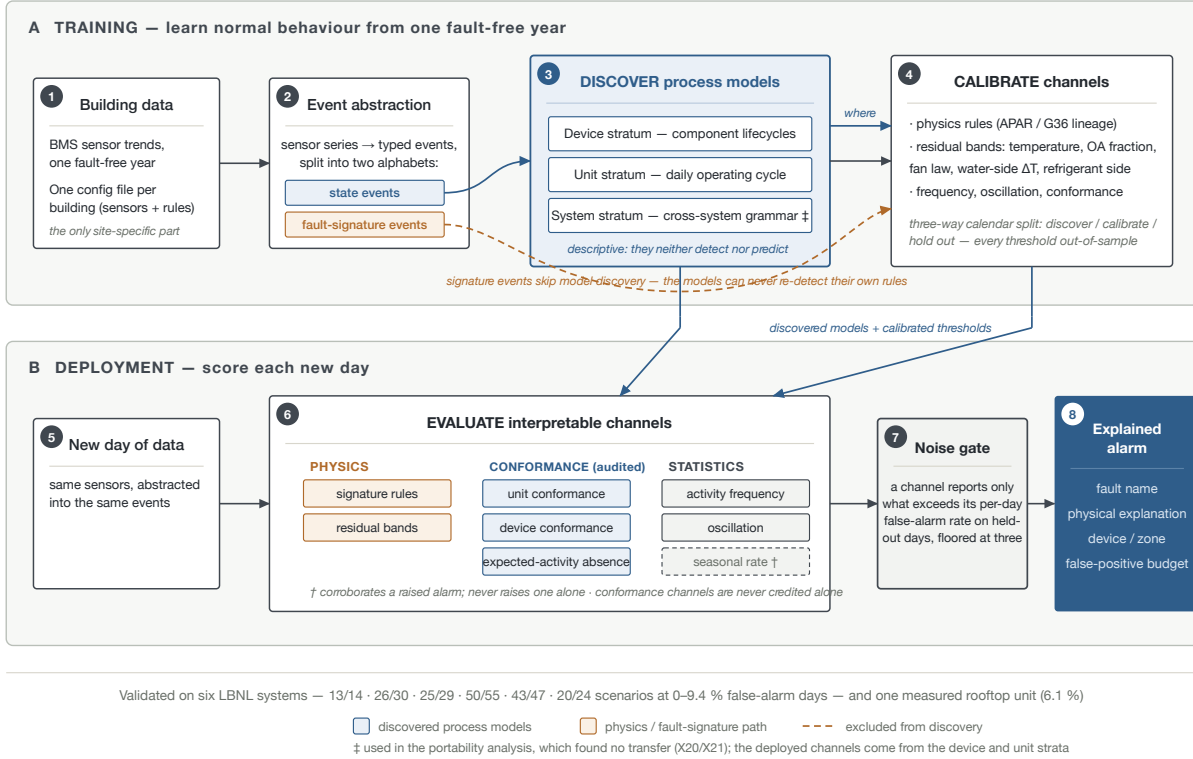


Figure 1: The framework as finally evaluated. Training (top): one fault-free year and one configuration file; events are split into state and fault-signature alphabets; process models are discovered from state events only and, on the evidence of Sections 6 and 6.2, describe normal operation and supply the strata and invariants the channels use, but contribute no detection of their own through conformance; every channel’s threshold comes from a three-way calendar split (discover, calibrate, hold out), so every reported false-alarm rate is out-of-sample. Deployment (bottom): each day is scored by physics, statistical and conformance channels; a channel reports only what exceeds its own per-day false-alarm rate on the held-out days it could evaluate, floored at three days; what survives reaches the operator with a name, a physical explanation, the device, and the stated budget.

4 Method

STRATA (Figure 1; code and artefacts at <https://github.com/iyassh/strata>) consumes one fault-free year of ordinary building-management-system trend data together with a single configuration file naming the building’s sensors and control rules. That configuration file is the only site-specific artefact: no code change was required to onboard any system reported here.

The framework is stratified in two senses. Events are abstracted from the control sequence at device and unit granularity, so that every alarm is born with a location; and the channels that read the resulting log — absence, frequency and the two conformance channels — sit beside the physics rules, the residuals and the oscillation channel, which read the raw signals, each calibrated against its own noise. The process-mining structure is what localises and organises detection; conformance against the discovered nets is one channel among eight, and Section 6 measures what it contributes.

4.1 Event abstraction and the alphabet wall

Continuous sensor streams are converted into a discrete event log using control-sequence rules grounded in ASHRAE Guideline 36, following the abstraction methodology of Janssen and colleagues [24] and the taxonomy of event abstraction in van Zelst and colleagues [47]; the same bridge from continuous sensor streams to process-aware event logs is pursued, for IoT data generally, by Sensor2EventLog [31]. The events are then partitioned into two alphabets that are never mixed. State events are neutral facts about operation and are the only events admitted to model discovery. Fault-signature events already encode engineering suspicion and are withheld from discovery entirely.

The partition — the alphabet wall — exists because of circularity, the process-mining form of the data-leakage problem [53], whose closest published relative is a benchmark-validity audit in sensor-based fault diagnosis that found reported accuracy re-encoding the labelling rules [42]: a model discovered over an alphabet that already names faults will later “detect” those faults by recognising the project’s own rules. Comparing this pipeline’s partitioned artefact against the committed artefact of its pre-partition prototype on the single-duct system — same threshold, same 248-day train and 87-day holdout split, same 4,540 fault days, and an identical rules-channel output of 3,134 true positives, but not the same code version — the model channel’s apparent recall rises from 1.8% to 5.2% (81 days against 238) without the partition. The two artefacts agree on every combined-detector number (3,152 true positives, 5 false positives), so the partition changed the deployed detector’s output by zero days: every model-channel detection the leak supplied was already caught by the rules channel. A controlled same-code re-run is future work; the deployed scorecard, two code generations later, gives the same reading (86 fault days strictly worse than the worst fault-free holdout day, the min-robust count, of 4,891 evaluable fault days, 1.76%; the thresholded model channel flags none). That 3.5-percentage-point difference (5.24% against 1.78%) is a measured bound on model self-confirmation, and the separation is now enforced by a tested invariant: injecting signature events into a log cannot change a day’s fitness. The process-mining literature warns about this trap in the abstract; we are not aware of a prior quantification of it on sensor-derived logs.

4.2 Discovery at three strata

Petri nets are discovered from the state-event log with the inductive miner [26, 27], as implemented in pm4py [3], at three granularities: the device stratum models an individual actuator’s lifecycle, the unit stratum models the air handler’s daily operating cycle, and a cross-system grammar supports the transfer analysis of Section 7.

Whether a discovered net encodes temporal order at all — rather than merely a set of permitted activities — is the question precision metrics address [7, 33, 43], and the imprecision of precision on logs without negative examples [48] is why we use a direct probe instead. The reversal probe is that direct check: every trace is reversed and re-scored, and the fitness drop is reported as a single number. The three models separate cleanly. The single-duct net falls from 0.8861 to 0.5526 under reversal, confirming genuine order sensitivity. The series fan-powered net scores identically to four decimal places (0.9533 against 0.9533) and the parallel net shifts by 0.0005 (0.9021 against 0.9026), showing that neither carries meaningful ordering information in its fitness. The probe is cheap, decisive and reusable, and it converted an informal claim about alphabet complexity into a measured fact.

4.3 Calibrated channels and the noise gate

Eight channels (Table 1) evaluate each day independently: (1) rules, physics rules descended from the NIST APAR lineage that name direct contradictions between measurements; (2) residual, temperature-residual bands encoding relationships that must hold under known operating modes — with the cooling coil inactive, supply-air minus mixed-air temperature should equal the supply-fan heat rise, so a biased sensor appears as a persistent offset; (3) model conformance and (4) device conformance, alignment cost [8, 46] against the discovered unit and device models respectively — these two are the process-mining channels; (5) absence, a scheduled activity that did not occur; (6) frequency and (7) oscillation, statistical bands on activity counts and reversals; and (8) a seasonal rate channel that corroborates but may not raise an alarm alone, for reasons disclosed in Section 9.

Calibration uses the fault-free year only, at a false-positive quantile of 0.01 against a calendar-stratified holdout of the last eight days of each month. No threshold may see fault data. Every rule additionally requires a positive control, because silence is not competence: a position-based leak rule proved structurally deaf on the fan-powered units, where the seeded leak passes water through a valve whose position reads closed honestly, and was replaced by a waterside flow rule.

A channel may then report only what exceeds its own measured noise floor, tested as a binomial at $p < 10^{-3}$ against that channel’s own false-alarm rate on the fault-free holdout days it could evaluate, floored at three days (a per-day rate for every channel since the statistical repair disclosed in Section 9). Detection thresholds are additionally floored at sensor physics: an early result on the series unit was retracted to at most two detection-days when a 0.5°F sensor-precision floor revealed that the “detections” had exceeded a 0.22°F band by 0.02 to 0.03°F .

What survives the gate reaches the operator with the implicated device or zone, the evidence that triggered it, and the channel’s own false-alarm budget. Figure 4, in Section 6, is one such alarm from the first day of a seeded fault, every value read from the committed artefacts and the raw data.

5 Verification as a First-Class Result

An audit of sixty-five primary machine-learning HVAC fault-detection studies found two that shared code links, one of them broken, and 72% that did not state whether their data was public, proprietary or commercial [32]. A field in that condition does not need another accuracy figure; it needs results that can be checked. We therefore treated verification as a deliverable rather than as hygiene, and this section describes the machinery in enough detail to be adopted. The refutations it produced are reported in Section 7.

5.1 Pre-registration and honoured falsifiers

Every experiment is pre-registered: the hypothesis, and the specific observation that would refute it, are committed to version control before the experiment runs, in the spirit of the hypothesis-testing discipline recently urged for process-mining research [37], of the pre-registration template proposed for predictive modelling [22], and of the pre-registered protocols the simulation-study literature recommends against spurious claims of superiority [36]. A falsifier that fires is treated as a finding and published, not as a setback to be absorbed by rewriting the hypothesis. Ten results went against the project — four refuted hypotheses, one fault-informed rule whose falsifier fired, three withdrawn claims, one pre-registered test that never had attainable power, and one four-building transfer test whose pass a discovery-free control reproduced — several of them results the project would have preferred to keep. Section 7 reports each with the claim it killed and what survived.

Four later pre-registered tests — component attribution, the diagnosis layer, quantile bands and the Guideline 36 comparison — also fired their falsifiers, and each is reported beside the result it bounds (Sections 6, 9 and 9.1). Appendix A lists every pre-registered experiment with its prediction and outcome.

The discipline also catches drift between the scripts that produce the published artefacts and the importable library: a pre-registered robustness experiment scored through the library detected five dual-duct scenarios the scorecard does not, under every condition including one that touches only the occupancy signal; the library’s residual noise floor had not received the per-day repair disclosed in Section 9, a defect the regression gate could not see because it regenerates through the scripts, and the one facade test of the time checked a single system’s per-channel flag counts, not the null pair. The library was repaired, a guard now pins its residual null to the scorecard’s, and the experiment was re-run against a clean arm through the same code path.²

The discipline exists because of a failure mode we observed in ourselves. One phase report, drafted the same day as its discovery, contained six distinct overclaims; all six were caught by audit within hours. Positive results attract inflated framing faster than negative ones attract scrutiny, and a falsifier is only protective if it is written down before the result it would kill is known.

5.2 Gates before science

Before any detection result is computed, a battery of gates runs against the raw data. It verifies file integrity by MD5, checks that timestamps increase monotonically, confirms that each file’s calendar is internally consistent, checks the raw files for calendar rotation, checks coverage of the Brick semantic model, and hashes each derived event log so that duplicate scenarios cannot masquerade as independent evidence. A separate healthy-silence gate is run per configuration, confirming that the rule set is silent on the fault-free year before it is permitted to score anything.

These gates were not designed in advance; each was added after a dataset defect got past the previous set. Three of the eight errata in Section 8 were found by this battery rather than by inspection — E1, E2 and E6 by the integrity hash. E4 was found by an early-phase audit; the calendar-rotation gate that now evidences it on the raw files was built four phases later, because the monotonicity check runs on converted files that conversion has already sorted and so could never have caught it. E3 was found by auditing our own baseline and E5 by a later hostile audit, which is the argument for running it: the defects are not visible in summary statistics, and every one of them would have inflated a published result.

5.3 Three levels of regression

Guards run at three levels. At the first, more than a hundred automated tests execute on every commit through continuous integration; the data-dependent tests skip rather than error when the datasets are absent, so the suite is meaningful on a bare clone. At the second, a regeneration gate re-derives committed artefacts from source and diffs each against its stored copy, reporting per artefact either an identical verdict or a named set of changed values. A changed verdict is a defect or a documented decision, never a silent update. At the third, every newly validated dataset is frozen permanently into the suite as a fixture, so that the examination the method must pass only ever grows.

The single-duct system is the byte-reproducibility standard: re-running its benchmark must reproduce the committed artefact byte-identically, and that requirement is enforced after every

²X31 Amendment 1, docs/plans/2026-09-24-x31-field-conditions-prereg.md; tests/test_facade_residual_null.py.

change. The fan-powered-unit artefacts are regenerated and diffed but carry no equivalent standing guarantee.

5.4 Tests that pin refutations and disclosures

The unusual property of the suite is what it guards. Alongside conventional unit tests of the event, residual, frequency and grammar logic, it pins three things that are not ordinarily thought of as testable.

It pins published values, so that a change which would alter a number quoted in a paper fails the build rather than propagating silently. It pins fired falsifiers: the sampling-rate experiment’s finding is its four refutations, and a regeneration that loses them is treated as a regression exactly as a lost feature would be. And it pins disclosure obligations. One test asserts that the artefact continues to declare which channels’ false-positive rows are calibration targets rather than out-of-sample measurements — the caveat reproduced in Section 9. Another asserts that the post-selection defence ships alongside the result it defends. A third recomputes the cry-wolf ratio from the benchmark and union artefacts rather than reading the stored value, so that an internally consistent but wrong number cannot survive; a fourth recomputes sensor coverage from the configuration files for the same reason.

Continuous integration has been used to guard that a published result regenerates [25]. What we have not found reported is CI that guards that a published caveat survives: a regeneration that drops a disclosure fails the build exactly as a lost feature would. Its justification is that caveats decay faster than code: a sentence in a document can be dropped in a rewrite without anything failing, whereas a test that enforces the sentence cannot.

5.5 Adversarial audit

By adversarial audit we mean an independent recomputation of a claimed quantity from raw data by someone briefed to find it wrong — in this project, AI coding agents given that brief and, for the sealed test of Section 7, denied access to the answer key — not the red-teaming of a model that the phrase denotes elsewhere. Before any number is quoted, it is attacked in this sense. An audit is briefed adversarially — to assume the result is wrong and to find the ways its authors have flattered themselves — and independently recomputes the quantity from raw data rather than re-running the pipeline that produced it, so that a defect in the pipeline cannot confirm itself. Independence of implementation is the point: on more than one occasion the recomputation agreed with the artefact and disagreed with the prose written about it.

The method’s value is measurable in what it caught. It found that a baseline was detecting file provenance rather than faults. It found that one of the fourteen single-duct detections was an artefact of the dataset — the fault-free file was simulated on a different configuration branch than every fault file, and the detector was noticing the branch, not the fault — so the score is reported as 14/14 naive and 13/14 adjudicated. It found that a day-level significance test adopted in Phase 4 rested on an impossible null, the discordant cell it tested being zero by construction; the test was retracted, the significance it had awarded was withdrawn, and a binomial against each channel’s own noise replaced it. It found detections below sensor precision, an evidence sentence falsified by its own recomputation, and, twice, a completed comparison against commercial practice: the first execution invalid for three independent reasons, the repaired one for six adapter defects and one defect of design (Section 9.1). None of these would have been caught by a test suite, because in each case the code was doing what it was written to do.

5.6 The lesson ledger, and why documentation decays

Sixty-seven numbered lessons record each failure with the standing rule it produced. The ledger is itself a result, because a pattern emerges from it that we did not anticipate.

The first pattern is recurrence: the same lesson reappeared in different clothing. A rule requiring day universes to be derived from the raw calendar rather than from the event log was recorded, documented, and then violated again two phases later. The conclusion drawn, and now standing policy, is that a lesson is not closed until a helper or a test mechanically enforces it — documentation alone decays. Several of the tests described above exist for precisely that reason.

The second is that every new detection channel recreated the same gap. Each needed its own noise-floor gate, and each time there was a temptation to exempt it; the rules channel was silently exempt until an audit found it. The harness now enforces the gate structurally rather than by convention.

The third is that simulation flatters. Noise-free simulated years make zero-false-positive thresholds cheap, bands razor-thin and silence easy, and a substantial share of the ledger reduces to that observation — which is why the sensor-precision floor and the noise-floor significance requirement exist at all, and why Section 9 treats the thinness of field data as the principal threat to these results. One measured stream now exists: on an LBNL field rooftop unit (Site 2; 182 fault-free days over two summers, 25 refrigerant- and air-side points, no schedule point), onboarded by configuration plus one documented derived fan-on column and two logged healthy-silence iterations, the deployed detector’s false-alarm rate is 3 of 49 held-out days (6.1 %) with every channel inside budget; its one fault case (a 40 % undercharge on one refrigerant circuit) is not seen, and is reported as a case, not a rate.³

5.7 Reproducibility

The datasets are public and citable by DOI [19]. The environment is pinned by lockfile, random seeds are fixed wherever randomness exists, and the repository documents the complete path from a fresh clone through data conversion, the gate battery, and the result pipeline to regenerated artefacts, with approximate wall-clock times. A claim-to-artifact table maps each published claim family — scorecards, joint false-alarm rates, baselines, the discovery-automation ablation, the grammar and reversal probe, the zone ground truth, the circularity bound, sensor coverage, cry-wolf ratios, the four robustness experiments, the alarm attribution, and the sealed transfer test — onto the specific artefact that backs it, so that a reader wishing to check a single number does not have to regenerate everything to find where it comes from.

One provenance note illustrates the standard we tried to hold. The circularity bound of Section 4 compares the current pipeline against its pre-split predecessor, which lives in a different repository. Rather than cite that repository, its artefact is committed here, so that both sides of the comparison regenerate from this one.

6 Results

Under the final gate STRATA detects 158 of 175 seeded scenarios on the five simulated systems (14 of 14 naive on the single-duct unit, 13 adjudicated; totals throughout use the naive count, so 157 under the adjudication) at 26 false-alarm days in 480 held-out fault-free days, at a median of one day, and names the implicated terminal unit in 37 of the 50 attributable detections with none

³outputs/x26_real_data_budget.json; docs/plans/2026-09-24-x26-real-data-budget-prereq.md.

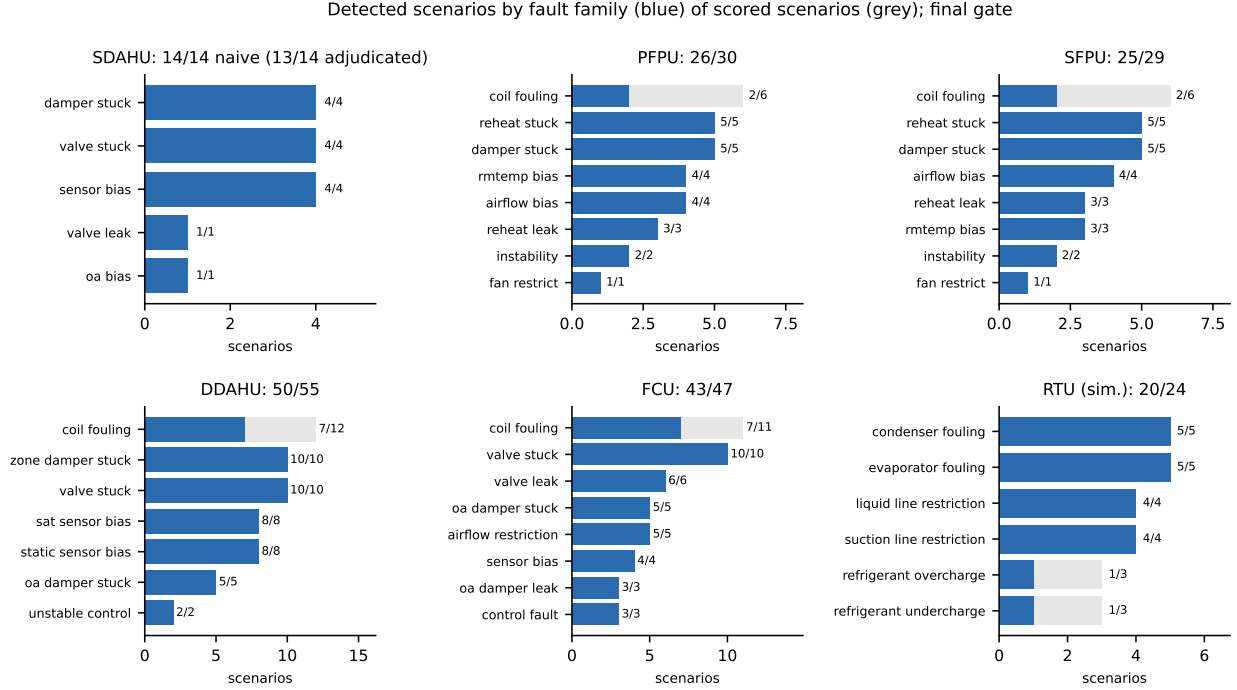


Figure 2: Detected scenarios (blue) of scored scenarios (grey) by fault family on the six scored systems, under the final gate. Coil fouling is the dominant miss on the four air-side datasets and is caught in full on the simulated rooftop unit, condenser fouling through the refrigerant side and evaporator fouling through the air side (Section 7).

wrong (Tables 2 and 3). What follows reports how, which channels carry it, what the process-mining channels contribute, and what remains missed.

Evaluation began with three public LBNL fault-detection datasets [18, 19]: a single-duct air-handling unit (SDAHU) and parallel and series fan-powered-unit configurations (PFPU, SFPU); four further datasets were onboarded afterwards (the dual-duct and fan coil units in Section 6.3; the two rooftop units in Sections 5 and 7), and Table 3 lists all seven datasets with Figures 2 and 3 summarising detections and false alarms on the six scored ones. The two fan-powered-unit datasets had not previously been benchmarked in the published literature: the searches described in Section 3 returned no published fault-detection result on either. The fan coil archive, by contrast, has since been used by an autoencoder study [28] and by a benchmark-validity audit [42]; no published detection result on the dual-duct unit was found. Scoring follows the day-level, scenario-as-unit conventions of the FDD evaluation protocol literature [14, 55].

The median time to detection is one day on all six systems, but the median alone misrepresents the distribution and should not be quoted without its tail (Table 4).

Gradually developing faults are detected slowly, which is the expected cost of calibrating below a noise floor and is a genuine limitation for the degradation faults that motivate much of this literature.

Sensor-bias detection illustrates why pooled recall misleads for gated detectors. Raw recall of 39–47% over the 351 to 352 evaluable days appears to be a capability ceiling until it is decomposed. The residual is physics-bound only inside a detection window — cooling coil off, zone occupied, sustained at least 120 minutes — and such windows exist on 137 to 164 days of the year in this

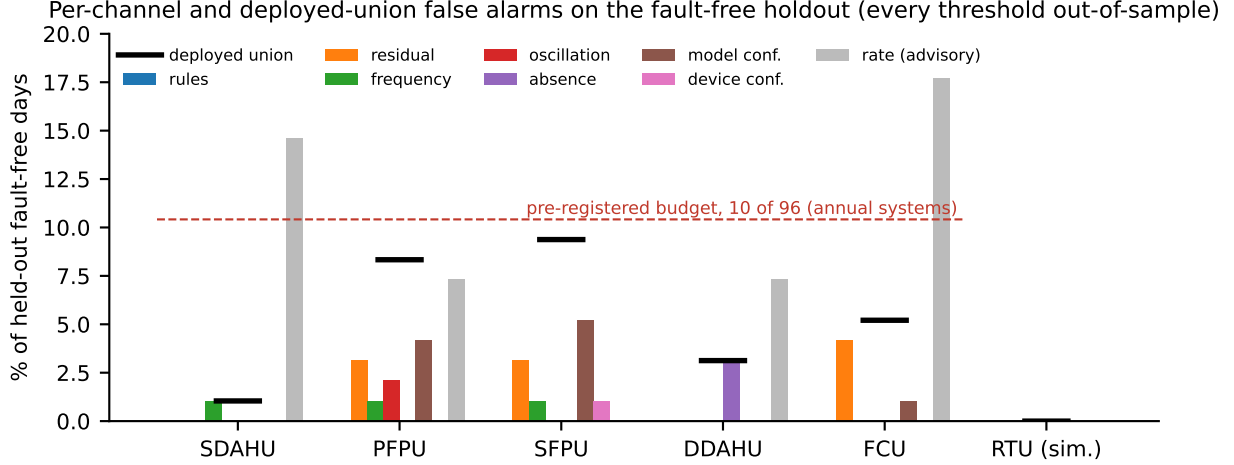


Figure 3: Per-channel and deployed-union false alarms on the held-out fault-free days, every threshold out-of-sample. The seasonal rate channel (hatched) is advisory and outside the deployed union; the dashed line is the pre-registered budget of 10 of 96 held-out days for the five annual systems; the simulated rooftop unit’s 28-day holdout carries no such budget line.

climate. Within them, window-conditional recall is measured at 100%, on counts of 164/164, 164/164, 139/139 and 137/137 across the four coil-bias scenarios. The apparent ceiling is climate, not capability, and reporting window prevalence separately from window-conditional recall is a methodological recommendation we would make to others evaluating gated detectors.

Against a principal-component-analysis baseline using squared prediction error, audited and scored on comparable scenario universes, STRATA detects three more scenarios than the baseline inside overlapping intervals (Table 5).

The misses are complementary rather than nested: three scenarios are detected only by STRATA and none only by the baseline. STRATA alone detects an unstable VAV damper and the parallel unit’s moderate and severe airside reheat fouling. Before the coverage audit of Section 6.4 the split was one against six, the baseline reaching three room-temperature biases, two series-unit airside fouling files and the series unit’s fan-flow restriction through zone-fan columns that STRATA’s configuration did not then map; every one of those six is now detected. Nine scenarios are missed by both. Eight of them are one family: reheat-coil fouling, waterside at every severity on both units and airside minor on both, is invisible to both methods; the ninth is the single-duct outdoor-air-bias run, a miss only under the branch adjudication of Section 8. That is the most useful negative result in the comparison. On false alarms the baseline is better on the two terminal-unit systems and within one day of STRATA on the air handler, on a different day universe; the intervals in Table 5 overlap throughout. On detection, 64 of 73 is [78.2, 93.4]% and 61 of 73 is [73.4, 90.3]% (Wilson), overlapping; an exact McNemar test on the three discordant scenarios gives $p = 0.25$, which with three pairs is an absence of power rather than evidence of superiority.⁴ Nor are the inputs symmetric: the baseline consumes 108 and 424 raw daily features on the single-duct and fan-powered systems where STRATA’s configurations mapped 12 of 30 and 56 of 109 columns when the comparison was first run and map 26 and 109 since the coverage audit (Table 5), so the inputs are now comparable in breadth but not in kind: the baseline consumes raw columns, STRATA

⁴Every interval and the paired test regenerate from the committed artefacts: `scripts/intervals.py` $\rightarrow$ `outputs/intervals.json`, under the regression gate.

STRATA alarm -- series fan-powered unit, 2018-01-01 (day 1 of the fault)
 what zone primary-air flow not tracking its setpoint [channel: rules]
 where terminal unit TU_S, zone S
 evidence measured 715 CFM against a 200 CFM setpoint -- a 515 CFM excess
 sustained for 720 min while occupied (rule fires at 50 CFM held
 60 min); damper position 0.80, fault-free same-day value 0.32
 budget rules channel: 0 false-alarm days on the 96-day fault-free
 holdout; threshold set on training days only, holdout out-of-sample
 corrob. seasonal rate channel agrees (advisory; never alarms alone)
 seeded zone-S damper stuck at 80% -- location correct, symptom correct,
 detected on day 1 of 365

Figure 4: A worked alarm. Scenario SFPV_VAVDMPRStuck_80%; rule zone_flow_tracking_S, a setpoint-deviation rule gated on occupancy. On day 3 the unit-model conformance channel also flags; this is the scenario on which a one-line rule later out-covered that model (Section 7). The alarm names the symptom and its location, not the root cause; a later pre-registered layer that maps signed residual patterns to fault families names the family in three of four resolved cases and is not yet adoptable (Section 6).

physically named relations between them. What the baseline does not do is name the implicated device or state the evidence behind a flag. That is the difference that matters operationally; Figure 4 shows it on one seeded fault, and it is the property a scalar score cannot produce.

The matched-rule control isolates what discovery contributes. For each channel a hand-written equivalent was authored and scored. Two of these reproduce their channel exactly — 141 detections against 141 on the series unit and 206 against 206 on the parallel unit — which is the point: the stratified log’s own count bands make explicit rules an expert could have written, had they known where to look (the matches are against a hand-written per-zone heating-episode band, located by the strata, not by the discovered net). Transplanted without that knowledge the same rules are unusable, one of them firing on 231 fault-free days on the single-duct system and 124 on the parallel unit while firing on none at all on the series unit — but the single-duct figure is not evidence about the models: that system has no heating rule and no hot-water valve, so a rule reading ‘heating never active’ fires on every occupied day for want of a sensor, not because any model predicted it. The only informative contrast is the parallel unit against the series unit, and whether the discovered models predict that contrast is the subject of a pre-registered test reported in Section 7. Some faults, conversely, need the classical instrument: a calibrated oscillation channel stands alone on the unstable-damper scenario, 349 detections against at most two from any matched rule.

The decisive ablation is the one the matched-rule control implies. Reading the significance-gated channel credited with each detection from the committed scorecards, no scenario on any system is detected only by the two conformance channels: they are credited with none of the 14 naive (13 adjudicated) single-duct or 26 parallel-unit detections, and appear in 7 of the 25 series-unit detections but never alone (counts after the coverage audit of Section 6.4, which left the model and device rows identical). Removing them leaves every scorecard unchanged. On the series unit they appear on 6 of the 9 deployed false-alarm days (out-of-sample since the three-way split); removing them eliminates five (the sixth is also flagged by another channel), lowering that system’s rate from 9/96 to 4/96 at no detection cost, and takes the parallel unit from 8/96 to 6/96 and the fan coil unit from 5/96 to 4/96 (recomputed from the per-channel day lists in the union artefacts). Whatever the discovered models contribute, they do not contribute detections.

Every alarm carries a location, and the location is right wherever it can be checked. On the 51

detections on the two fan-powered systems in the attribution universe (65 naive detections less the 14 on the single-duct system, which has no device stratum; one with indeterminate ground truth), the alarm names the correct terminal unit in 37, names none in 13, and names a wrong unit in none — three wrong units were available in every case, so the zero is falsifiable even though the 37 is not. Localisation is reported as specificity rather than discrimination: because every seeded zone fault in the benchmark occurs in the same zone, the ground truth has no variance and a perfect score would be unfalsifiable as evidence of discriminating between zones, so no per-zone discrimination figure is quoted. A second, pre-registered attribution test asks where in the equipment the alarm points: over all 158 detections on the five simulated systems, the rule carrying the detection names the seeded component exactly in 109 (69%) and the right subsystem but a neighbouring component in 22 (14%) — 83% at subsystem level — the wrong subsystem in 20 (12.7%), and no component in 7 (four carried by the frequency or oscillation channel alone, three by residual rules the attribution table does not name). The pre-registered bar for a component-level claim (wrong subsystem $\leq 10\%$) was missed, so the claim is subsystem-level. Fourteen of the twenty wrong cases are on the single-zone fan coil unit and name the component that responds to the fault rather than the one that caused it: the coil for a biased room sensor, the fan for a fouled coil, the valve for a reversed controller — the first place a technician would look, but not the cause.⁵ A pre-registered diagnosis layer built from three fault-agnostic ingredients — the sign of every significant residual against its healthy band, a consensus rule over redundant copies of one quantity (the four mixing-box statics of a deck, the four return-minus-zone residuals), and a leave-one-out nearest-pattern vote for the fault family — raised the exact-component count to 112 of 158 (70.9%), 85% at subsystem level, and lowered the wrong-subsystem rate to 10.1%: the consensus rule resolved the five dual-duct damper cases the carrying rule had scored wrong (one box copy moves alone) and called 14 of 16 deck sensor biases at the deck (all four copies move the same way), which is the mechanism the pre-registration described, working where the redundancy is real. The layer did not clear its bars: the family is named from the signed pattern alone in 85 of 114 resolved detections (74.6%, bar 80) with 44 unresolved (27.8%, bar 15), and the wrong-subsystem rate of 10.1% missed the bar (below 10) by one case. It does nothing on the fan-powered units, where a single zone’s fault moves that zone’s return-minus-zone residual one way and the other three the opposite way because the return is the mix; and the family confusions that remain — a stuck damper against a biased airflow sensor, a restricted filter against a fouled coil, a leaking against a stuck valve — are pairs the recorded quantities cannot separate by sign.⁶

Portability was measured. The second building type was onboarded through configuration alone in approximately 1.5 hours of wall-clock time, reconstructed from commit timestamps, comprising 45 sensor mappings and 35 rules at onboarding (109 mappings and 58 rules in the committed configuration, after the coverage audit of Section 6.4); the third was derived from the second in minutes. No pipeline code changed.

6.1 What the event log carries, and the time perspective

The null of Section 6 was argued from the model side — order-permissive nets, count deviations surfacing as alignment cost — until the log itself was examined. For the parallel fan-powered unit we compared, day by day, the fault-free year against a missed scenario and a detected one on the state alphabet alone. Under waterside reheat-coil fouling at its most severe the state-event log is identical to healthy to the minute: 162.8 against 162.4 events per day, the same

⁵outputs/component_attribution.json; docs/plans/2026-09-25-x34-component-attribution-prereg.md.

⁶outputs/x37_diagnosis.json; docs/plans/2026-09-27-x37-diagnosis-layer-prereg.md; three runs, two script defects corrected to the pre-registered wording, earlier runs kept.

activity counts, a zone-S heating-on median of 124 minutes in both. Over all 73 scenarios, at a 5% tolerance on every activity’s daily count and every state’s on-duration median, eight of the twelve scenarios the deployed detector missed at the time (sixteen under the final gate) have a state log indistinguishable from healthy — all reheat-coil fouling — and for them no method that reads that log can detect, process mining or otherwise, because at the deployed alphabet the fault lives in heat-transfer magnitudes the abstraction discards; the other four change a daily count by 7–39% that the frequency channel’s bands did not resolve into significant days.⁷ Under a 50% reheat-valve leak the log does change — the zone-S heating-on median falls from 124 to 11 minutes and the zone fan cycles 26.6 times a day against 16.3 — in timing and in counts, with the activity set and its order unchanged. Control-flow conformance is blind to timing by construction; the frequency channel already reads counts.

That diagnosis fixed a pre-registered test of the one perspective never tried (X12): a time-infused state model — for every on/off pair of the state alphabet, the day’s median on-duration, off-gap and first-on minute, train min/max bands widened by five minutes, the holdout false-alarm floor and the same binomial gate as every channel. Against the committed scorecards’ flag days it is significant on 11 of 73 scenarios with p down to 10^{-94} (10 under the more conservative rule-of-three floor), on none of the twelve fouling scenarios (predicted), and on no scenario the deployed channels miss (the load-bearing prediction, refuted); it is never earlier than the deployed detector (typically day 14 against day 1) and it costs five holdout false-alarm days on each fan-powered system, taking the parallel unit’s then-deployed joint rate from 5.2% to 10.4%, over budget. What it adds is a small day-level coverage supplement on scenarios already caught — 81 unique days on the single-duct coil-bias run (15 against the gated union), 16 on a series-unit sensor-bias run, none on five further series-unit scenarios, four reheat-valve faults and a stuck VAV damper, that the rules already flag on every day — through physically legible statistics such as zone-heating on/off medians and the cooling first-on minute.⁸

A fault-informed coil-effectiveness residual for the fouling misses (X13, design disclosed as chosen after inspecting three fouling files, thresholds from healthy train days only) then fired its falsifier: one of twelve fouling scenarios significant, nothing newly detected. A last pre-registered variant asked whether the model had simply been starved of vocabulary (X14): healthy-derived band states for every actuator position and flow (above the occupied 90th percentile, at or below the 10th, fifteen-minute dwell; the wall intact) enrich the alphabet by 6, 40 and 36 events. The single-duct model channel then fires on all 14 scenarios instead of none, and one of the twelve misses, moderate airside fouling of a series-unit reheat coil, becomes significant on the frequency and time channels — at a holdout false-alarm rate of 7, 18 and 21 days in 96 (on the calendar-day universe), over the budget on both fan-powered units, where alignment-based conformance on the enriched log was also computationally impractical (single-laptop timings, printed and not committed: seventeen minutes to fit; one scenario not scored in forty-seven) and was not evaluated; X14’s falsifier fired.⁹ The detection is carried by one added state, the zone damper’s low band, entered nine times a day against under one on the healthy year; a committed fault-versus-fault control over all 31 series-unit files shows a monotone ladder (64%, 64%, 12% and 3.5% of occupied samples at or below the band edge from healthy to severe), with severe firing on the same statistic and already detected, and the three waterside-fouling files sitting exactly at the healthy values at every severity, so it is a physically legible, dose-monotone signature of airside fouling that the deployed alphabet did not

⁷outputs/x12_log_diagnosis.json.

⁸outputs/x12_time_perspective.json; pre-registration and fourth prediction’s weakness disclosed in PHASE9_RESULTS.md.

⁹outputs/x14_enriched_alphabet.json; three amendments, each recorded before the result it governs, in the pre-registration.

carry — on channels that false-alarm one clean day in five.

A post-hoc read of the same artefact, not a pre-registered result, locates the cost in the time model (15 and 19 false-alarm days in 96) and finds the frequency channel alone, on the enriched alphabet, at 4 and 3 in 96 and still carrying the detection. That read was then pre-registered as a test with its provenance stated and the holdout as the thing that could still falsify it (X15): the enriched frequency channel alone stays inside the budget under both the deployed and the mirrored eight-day holdout (4 and 8, 3 and 5 days in 96 on the fan-powered units), detects the airside-moderate fouling scenario under both, adds nothing else new — a scenario the principal-component baseline also detects from raw features (Section 6), so the gain is within reach of a simpler method and the claim is only that an enriched event log carries it too — and is significant on 41 of the 61 scenarios the naive scorecard already detects against the deployed frequency channel’s 12 — at three to eight times that channel’s own false alarms (1 in 96 on every system), which is the like-for-like price of the coverage. No falsifier fired. It is the one constructive result of the night: a candidate deployable addition, not an adopted one, since adoption changes every published scorecard and is a separate decision, to be pre-registered.¹⁰ The reading is now measured rather than argued: process mining’s contribution on these systems is the event log and its count and time perspectives; the discovered net’s order structure contributes nothing at the deployed vocabulary, and richer vocabulary buys detections at false alarms the budget forbids.

6.2 What the conformance channel can see: a positive control

The null above can be read two ways: the faults do not change event order, or the channel cannot see order. A pre-registered positive control (Figure 5) injected order violations — whole-day reversal, adjacent swaps, deletions, a deleted start event, a duplicated block — into each system’s fault-free holdout traces and scored them with the deployed net, threshold and significance gate, adding two threshold-free measures: the AUC of perturbed against unperturbed per-day fitness, and the hit rate of a threshold set at the fault-free holdout’s worst day. The channel turned out to be three instruments. On the fan coil and dual-duct units a reversed day is flagged on 55 of 72 and 59 of 83 holdout days (AUC 0.99; 75 and 71 % hit rate at zero false alarms): the channel sees order, and those units’ faults do not change it (the dual-duct unit joined this group only once its threshold was calibrated out-of-sample, Section 9). On the single-duct unit a reversed day is separable on average (AUC 0.91) but the hit rate at zero false alarms is exactly zero: the fault-free holdout contains days that fit worse than a reversed day, so no threshold inside the budget can see order — blind by the spread of healthy fitness, not by the net. On the two fan-powered units fitness does not move under any perturbation (AUC 0.36–0.51; deleting the start event raises it): blind by net structure. Eight adjacent swaps and four deletions are also significant on the two order-seeing units, consistent with alignment cost behaving as a graded detector there. No falsifier fired, and four of the five predictions were wrong.¹¹

The instrument was then varied. A pre-registered grid of seven process-mining instruments — the inductive miner at noise 0.0 and 0.5 with alignment fitness, the deployed net with token-replay fitness, the heuristics miner with alignments, the log skeleton, Declare constraints and the temporal profile — was discovered on each system’s training days, thresholded at the 1% quantile of the per-day score on the calibration slice and scored through the deployed significance gate on the fault-free holdout and on the 33 scenarios the deployed detector then missed (8, 8, 10 and 7 on the fan-powered, dual-duct and fan coil units; sixteen of them have since been recovered by the

¹⁰outputs/x15_enriched_frequency.json; the control in outputs/x14_control.json; per-channel false-alarm attribution in outputs/x14_enriched_alphabet.json.

¹¹outputs/x22_positive_control.json; docs/plans/2026-09-24-x22-conformance-positive-control-prereg.md.

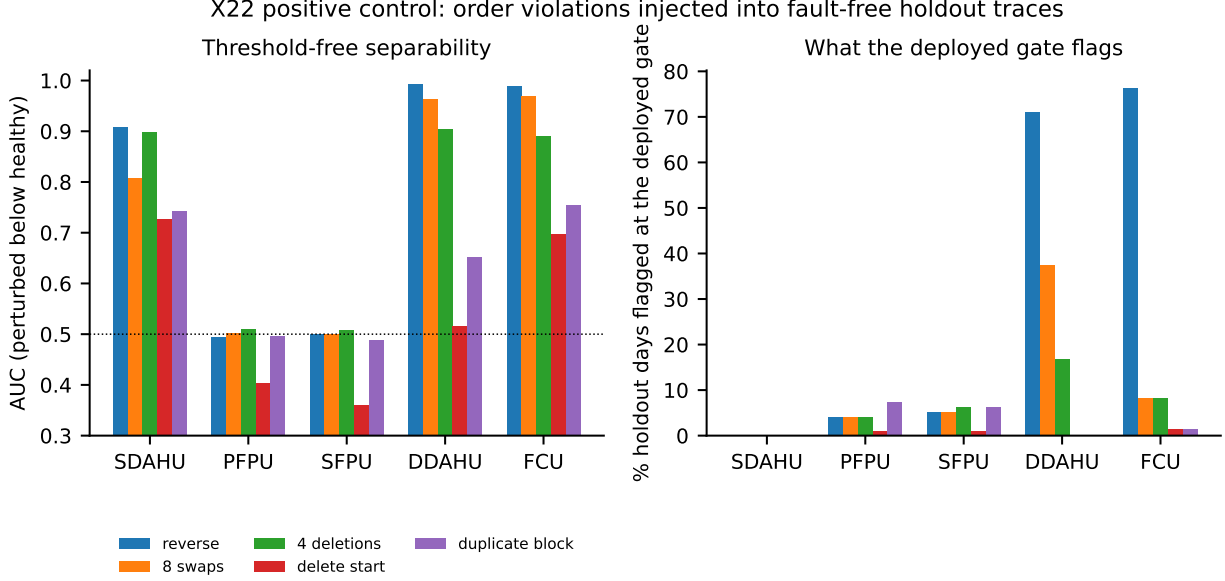


Figure 5: The positive control. Left: threshold-free separability (AUC) of perturbed against unperturbed fault-free holdout days for each injected order violation. Right: the share of those days the deployed gate flags. The fan coil and dual-duct units see order; the single-duct unit separates it on average but not at any in-budget threshold; the fan-powered units’ nets accept any order.

coverage audit of Section 6.4, through columns no process model was ever given). No cell detects any of them at its threshold (realised false-alarm rates 0 to 9.7%; no threshold sweep was run); the highest share of scenario days any cell flags is 5.6%, by the log skeleton on a severe water-side fouling file, at 7 false-alarm days in 72. No cell exceeds 10% false alarms, so the alternative instruments are silent rather than noisy. Two heuristics-miner cells exceeded a 15-minute budget and are reported as not evaluated, not as nulls. The conformance null is therefore a property of the state-event log, not of the miner or the measure: what the log does not carry, no conformance instrument over it recovers.¹²

6.3 A fourth and a fifth system, onboarded blind

The onboarding claim — that a building is a configuration directory, not code — was finally tested on a system type the project had never seen, under a pre-registration written after the documentation and before any data file was opened: the LBNL dual-duct AHU (two decks, two coils, two supply fans, three dampers, four zone mixing boxes with command-only dampers; 114 points; 55 seeded faults in 15 families, two of them — static-pressure sensor bias and unstable control sequences — absent from the first three systems). The configuration took 80 canonical sensor mappings (79 recorded columns, as Table 6 counts them, plus the timestamp), 15 state rules and 26 signature rules at onboarding (56 in the committed configuration, after the residuals added by later experiments and the coverage audit) reusing the existing rule kinds, and one healthy-silence iteration; no source change. That iteration is the instructive part: ten healthy days on which the cold-deck fan saturates and cannot hold its static setpoint drove nearly every healthy firing, and they were removed by a physical gate (duct static at setpoint) rather than a looser threshold. The

¹²outputs/x30_method_grid.json; docs/plans/2026-09-24-x30-method-grid-prereg.md.

gate battery, extended with a configuration-branch comparison of the kind that found E5, found no defect in the dataset.

The detector caught 45 of 55 scenarios at 3 false-alarm days in 96 (8 with the advisory rate channel counted; 50 of 55 after the coverage audit of Section 6.4, which brings the static-pressure and deck-temperature biases to eight of eight each and the coil fouling to seven of twelve): every zone-damper, outdoor-air-damper, coil-valve and unstable-control scenario, seven of eight deck-temperature biases, five of twelve coil-fouling scenarios (the pre-registration predicted at most four) and six of eight static-pressure biases (the sixth through the fan-law static residual added later). The discovered model channel is credited on two cold-deck damper scenarios that four other channels already carry (none before the three-way split); nothing was detected by the conformance channels alone. The enriched-alphabet frequency channel of X15, re-run here under its own protocol, stays inside the budget and triples its coverage of already-detected scenarios but catches none of the ten misses: the one fouling gain of X15 was a one-system signature and does not replicate. One check misfired usefully: the false-alarm script reported that demoting the rate channel would delay one scenario’s first alarm, and recomputation showed the day-one alarm belonged to the absence channel, whose day list the scorecards do not export — a blind spot in the check, now recorded as such, and no cost of demotion. Two things follow upstream: the transfer test’s scorable pool grows from two buildings to three (its amendment excluded the single-duct system structurally, for want of a heating activity), so the building-level floor moves from $1/2! = 0.5$ to $1/3! = 0.167$ with the dual-duct unit and to $1/4! = 0.042$ with the fan coil unit below — the first design that can attain $p < 0.05$; that re-run is row 10 of Table 7; and the dataset defects are properties of particular archives (every one of the five carries at least one, four of them on the single-duct unit), not of the collection.¹³

The fifth system, the LBNL fan coil unit (a four-pipe single-zone unit with a three-speed fan, both coils and one outdoor-air damper; 29 points; 48 seeded faults in 17 families, four of them — filter restriction, outdoor-air inlet blockage, fan outlet blockage and reverse-acting control, scored as two — absent from the first four), was onboarded under the same pre-registration form: 29 mappings, 7 state and 16 signature rules at onboarding (17 after the outdoor-air-fraction residual described below), two logged healthy-silence iterations, no change under src/ (the gate script needed two lines to read this system’s Brick file). The hash gate found a byte-identical pair under two family labels (cooling and heating airside fouling, minor); the content is scored once and the fouling family is 11, the scored total 47. The first run detected 47 of 47 at 28 false-alarm days in 96 and fired two falsifiers. Twenty-five of the 28 were the model channel’s, every one a Saturday or Sunday (the other three were residual days, and they survive into run 2): the two scoring scripts flag an event-less scheduled day as a model violation and define scheduled as any non-zero mode — a rule written for the fan-powered units, whose mode 2 is a night cycle that generates events — and on this unit mode 2 is idle setback, so 97 of 104 weekend days were “silent while scheduled.” The scorecard’s model gate then judged six scenarios significant against a conformance-only baseline (1 of 72 event days) that the deployed count (25 of 96) did not use; their 100 model days each were the healthy year’s own 100 weekend model days: 97 silent and three two-event traces. A two-script repair — scheduled means operate mode, the definition the occupancy event kind already used, and the gate’s baseline counts what the deployed channel counts — was pre-registered with four predictions and three falsifiers before it was made. All four held: the four earlier scorecards regenerated byte-identically, the fan coil unit scores 41 of 47 at 4 false-alarm days in 96 (residual 3, model 1), exactly the six model-only scenarios dropped and nothing else changed status, and

¹³outputs/x17_onboarding.json, benchmark_v6_ddahu.json, union_fpr_ddahu.json, week0_audit_ddahu.json; pre-registration and one amendment in docs/plans/2026-09-24-x17-ddahu-onboarding-prereg.md.

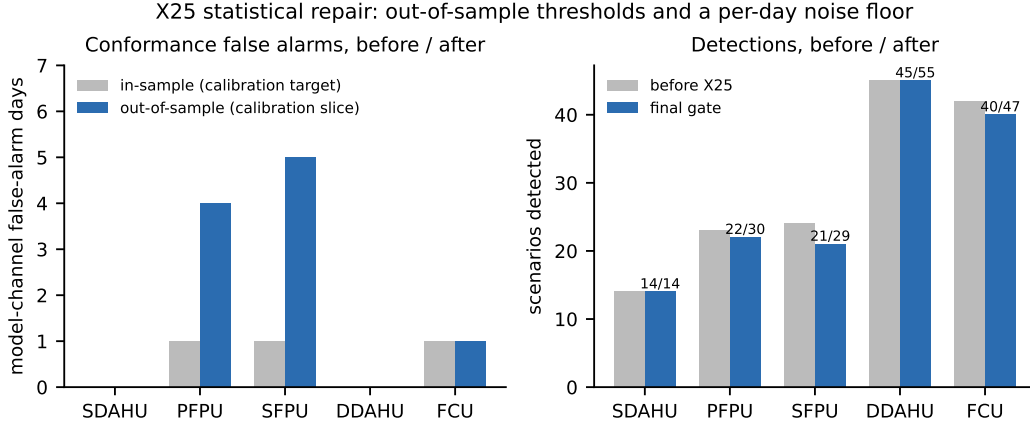


Figure 6: The statistical repair. Left: the conformance channel’s false-alarm days as a calibration target (grey) and out-of-sample after the three-way split (blue). Right: detections before and under the final gate; the per-day residual null removed seven residual-only detections, one offset by a new channel.

nothing is detected by conformance alone (a pre-registered outdoor-air-fraction residual, Guideline 36 FC6 lineage, later caught the 20 % damper leak at unchanged false alarms, and the per-day residual null of the statistical repair, Section 9, took it back again: 40 of 47 under the final gate); the model channel is credited on six scenarios under the final gate, each also caught by rules, residual, frequency or oscillation. Both new families are caught in full (airflow restriction 5 of 5, control 3 of 3); the misses at onboarding were five of six waterside-fouling files (four after the coverage audit of Section 6.4) — on the cooling coil the recorded water-side temperature difference, valve-open time and flow stay within about three per cent of the fault-free year — and the 20 % outdoor-air damper leak; the per-day residual null of the statistical repair (Figure 6) then took back both the 20 % leak, which the outdoor-air-fraction residual had caught, and the 50 % leak, which had been detected at onboarding, so both leaks were misses until the coverage audit recovered them.¹⁴ What the fifth system adds is not the row but the failure: a semantic constant in a scoring script is a hypothesis about the equipment class, and two artefacts that compute one quantity two ways will agree until the first class on which they do not.

6.4 The coverage audit: every recorded column, one system at a time

The misses that remained after the statistical repair were read as limits of what the datasets record. A read-only scan then found that for six of the fourteen misses it classed as reachable from a recorded column the decisive signal was a column no configuration mapped (the audit that followed recovered sixteen, several of them fouling files the scan had classed as unreachable): the configurations had been written from the control sequences, not from an audit of every recorded column. A pre-registered series (X33a–e) audited every column of every system in the project’s onboarding order, mapped every usable one, gave each a physically named reader — a residual, a rule, a gate or an oscillation signal — and excluded the rest with a written reason that a build-time guard now enforces. Every band came from the fault-free file; the state alphabet was not touched, so the process-mining rows had to regenerate byte-identically; and a pre-registered adoption rule

¹⁴outputs/x19_onboarding_run1.json, x19_onboarding.json, benchmark_v6_fcj.json, union_fpr_fcj.json; docs/plans/2026-09-24-x19-fcj-onboarding-prereg.md with Amendment 1; tests/test_artefact_consistency.py.

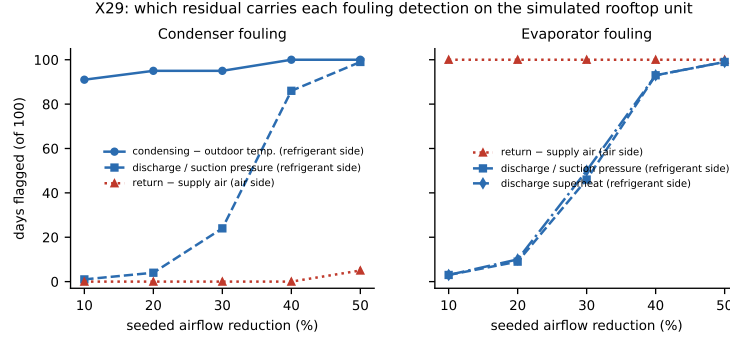


Figure 7: Which residual carries each fouling detection on the simulated rooftop unit, by seeded airflow reduction (days flagged of 100, from the per-residual credit artefact). Condenser fouling is carried by the refrigerant-side condensing-temperature approach at every severity; evaporator fouling is carried by the air-side return-minus-supply residual at every severity, with the refrigerant-side residuals joining from 30 %.

removed any new channel that was credited nowhere and cost more than one held-out day (none was).

What the recovered misses were, and which physics carried them (Table 6): airside reheat-coil fouling on both fan-powered units, through the series fan’s power per cfm of air delivered (“same power, less air”) and, on the parallel unit, the relationship between two airflow sensors; the fan-flow restriction, through the same fan-power witness; the room-temperature sensor biases, through the return air falling away from the biased zone’s reading; the hot-deck static and temperature biases on the dual-duct unit, through the unbiased copies of both quantities the mixing boxes record (exact, 282 of 282 days, at zero false alarms); one waterside fouling, through the pump’s bypass flow; and the two damper leaks on the fan coil unit, through the outdoor-air fraction of discharge flow at a fixed damper position. Two of the recoveries deserve a caveat the ledgers carry: the fan coil unit’s moderate heating-waterside fouling is seen only because the controller drives the fan to a speed the healthy year never used, an operating-point witness rather than a coil measurement; and one predicted recovery on the parallel unit’s waterside fouling is visible in the data yet uncertifiable, because the water flow only reveals it on the ten days a year the reheat valve saturates, too few to estimate a null.

Two findings of the series are about the protocol rather than the detector. The first step exposed a drift between the importable library and the benchmark scripts (Section 5); the second exposed that the fan-powered units’ documentation names the wrong zone for every seeded fault (erratum E7, Section 8). The seventeen misses that remain are coil fouling on every system, whose recorded values sit inside every healthy band on every column. The claim the series licenses is narrow: on these datasets, the remaining misses are limits of what is recorded and how often, not of the detector’s vocabulary or its gate — and the earlier reading, that all thirty-three were, was wrong for sixteen. Two disclosures bound the claim: which columns to read was decided after a read-only scan of the fault files had shown which columns separate which misses, so the readers were fault-informed even though every threshold was blind; and five of the sixteen recoveries sit less than one band width outside the healthy extremes, so a different healthy year could move the band edges they clear. Two later pre-registered tests bound that second disclosure: all five edge recoveries survive bands about ten per cent narrower, and four of the five survive injected field conditions (Section 9). The series therefore shows that these misses were reachable from the

recorded columns with blind thresholds, not that a blind onboarding would have reached them; the blind onboardings of Section 6.3 remain the honest prior for a new building.

7 What the Protocol Refuted or Withdrew

The detections and localisation of Section 6 stand on the stratified structure; this section reports what the protocol removed from the claims around it. Four pre-registered hypotheses were refuted during this work, one fault-informed rule fired its pre-registered falsifier, three of the project’s own results were withdrawn, one pre-registered test was found to have no power, and its powered re-run passed for a reason that has nothing to do with the hypothesis (Table 7). They are reported because a protocol that only confirms is not a protocol, and because these refutations constrain the contribution more usefully than the confirmations do.

Discovered models do not out-detect hand-written rules. On the series fan-powered unit, zone damper-stuck and negative airflow-bias faults erase the unit’s daily heating rhythm, and the discovered healthy net flags the absence. A one-line matched rule — occupied workday and AHU heating never active, with zero false alarms on 365 healthy days — covered 192 days against the model channel’s 135 on the same scenario (178 in the current artefact, after the threshold repairs; the falsifier fires either way). The 135 is a threshold-based count and should be read alongside the min-robust count — the count that survives replacing the interpolated threshold with the most conservative one — of 22 to 26 days per scenario, since roughly four fifths of the threshold count is attributable to a threshold interpolated from a single holdout day. The falsifier fired. An invariant surfaced during discovery and the rule was written afterwards; whether the model could have predicted where that rule would hold is the question the seventh entry could not test. The matched rule is now a permanent ablation arm. This refutation is why the paper does not present the models as a detector.

Behavioural structure does not transfer zero-shot. The cross-system grammar was pre-registered with the hypothesis that structure discovered on one building would support detection on another. The grammar analysis did find real shared structure — fan-powered-unit logs outscore their own shuffled controls on the single-duct net by 0.15 and 0.18 in mean fitness, while the other four cross-system cells are null (order signal between -0.005 and 0.000), so only the single-duct net is order-sensitive — but the detection hypothesis failed its pre-registered falsifier, yielding zero to one gated detection per rotation. Portability is therefore claimed at configuration level only, which is what the onboarding evidence supports and no more.

The models’ detections were count deviations in disguise. Audits of the device-stratum results reached one conclusion repeatedly: on loop-heavy alphabets the discovered nets are order-permissive, and every genuine model-channel detection was a frequency deviation — extra heating episodes, collapsed fan cycles, a vanished daily rhythm — surfacing through alignment cost. The frequency channel was built to measure those deviations directly and two-sidedly, and once it existed the conformance channels had nothing left that was uniquely theirs, which the ablation above confirms.

A pre-registered test that never had power. After the ablation one claim remained: that the models locate invariants, so that a rule written from them would hold where the model’s support for the invariant is high. A test was pre-registered — the model-derived quantity, predictions and falsifiers fixed and committed before computation, the scorer committed before data and then changed once after the model quantity existed (Monte Carlo to exact permutation p , a change that cannot alter the verdict but was made without a dated amendment), and the quantity computed by an agent forbidden from opening the rule-firing artefacts. The seal was narrower than that

sentence implies, and the pre-registration’s first amendment records why: the disclosure section of the pre-registration itself printed the transplanted rule’s healthy firings (231 / 124 / 0), and the agent was instructed to read it. Blindness therefore held only for the four cells belonging to two count-band rules whose bands were fitted on the same fault-free year they were scored on — cells whose near-zero firings are arithmetic, not transfer — while the two cells carrying information, those of the one parameter-free and genuinely transplanted rule, were unblinded by construction.

The test also never had power, for a reason neither the pre-registration nor its first two amendments caught. The model quantity is a property of a building’s net and is constant across rules within a building (0.000, 0.455 and 0.978), so the nine cells were three buildings pseudo-replicated; the valid permutation null is over buildings, and with three the minimum attainable one-sided p is $1/3! = 0.167$ — with the two buildings the amended test actually scored, $1/2! = 0.5$ — against a criterion of $p < 0.05$. A dated amendment made before computation excluded the single-duct system for a correct reason — it has no heating rule and no hot-water valve, so a rule reading ‘heating never active’ fires on every occupied day for want of a sensor, and its 231 firings are not evidence about any model, which also retracts the project’s earlier reading of that number — and a second amendment, made after the fact, diagnosed the resulting six-cell design as the cause of the power loss. That diagnosis was itself wrong: the cell-level design had never been valid. A third amendment records this. The reported $p = 0.20$ is not a measurement, the fired falsifiers license no conclusion, and the one cell the second amendment had kept as informative compares against a predictor the first amendment had already disqualified. The paper draws no inference about the claim in either direction.

What the exercise showed is reported as observation. The series unit’s net treats heating as obligatory — 356 of its 364 healthy daily variants contain the activity, and none of the 364 fits the net once it is deleted — while the parallel unit’s net treats it as optional; the alignment-based support equalled the raw log count exactly on every system, so at the unit stratum the net’s reading added nothing over counting; and at the device stratum the pooled net ordered the two buildings the opposite way. The lesson is the entry’s contribution: pre-registration fixed the quantity, the predictions and the falsifiers, and two amendments written under the same discipline by the same author did not fix the unit of analysis. A power calculation on the wrong unit survives pre-registration intact. The remedy adopted is a pre-registration field naming the unit of replication and the minimum attainable p under a permutation of that unit, computed before any prediction is written; a powered test of the transfer claim needs more buildings than three.

Calibration is fragile to contaminated training data. Extreme-value calibration has a breakdown point of one observation in n , and a single score-carrying contaminated day in the fault-free year is sufficient to collapse the two negative rungs of the coil-bias ladder from 164 detections out of 164 to none, by dragging the band edge to -7.19 . The finding must be quoted with its detectability split or it misleads in either direction: contamination from a mild source announces itself on 100% of the contaminated days (5/5, 13/13 and 27/27 signature days) and is therefore caught, whereas the dangerous source is visible on none of its 27 days. A related test found that configurations tuned at one-minute sampling gain 28 to 64 spurious healthy signature-days at fifteen-minute sampling; four falsifiers fired, and the transfer claim was scoped so that the healthy-silence gate is re-run at the deployment sampling rate rather than transplanted.

8 Defects in the Benchmark Datasets

Auditing the evaluation produced an unanticipated secondary contribution: eight defects in the LBNL benchmark datasets, each with machine-verifiable evidence and a reproducing command

(the creators’ own account of the models’ fidelity limits [9] is the closest prior; the defects here are byte-level and label-level findings, not fidelity findings) (Table 8). One of them, the configuration-branch mismatch described above (E5), hid for six phases because every comparison ran against the same fault-free file and no fault-versus-fault control existed.

The most consequential for other researchers concerns the supply-air static-pressure columns, which carry a per-branch additive constant of about 401.86 — added to one column in the fault-free file and subtracted from the other in every fault file — with no overlap between the two ranges. Either column alone separates fault-free from faulty data by file provenance rather than by fault physics: a single threshold on either column, chosen from the fault-free file alone, classifies all 365 fault-free days and all 7,150 days of the twenty fault files correctly (accuracy 1.0 on both columns), so any model given these columns can identify provenance perfectly without reference to any fault. We discovered this by auditing our own baseline, whose error statistic had reached implausible magnitudes for precisely this reason; the columns are now excluded from all baselines. Published results on this dataset that used the full column set should be read with this in mind.

9 Limitations

The false-alarm budget is the property that has been stressed hardest, and it held: under injected sensor error, change-of-value logging and schedule shifts on the shipped configurations of all five simulated systems the worst cell is 7 of 96 held-out days, each condition costs at most one detection per system, and a narrower calibration gains nothing, as the paragraphs below record. The limitations that remain are these.

All detection results come from simulation, and simulation flatters. Two pre-registered injections, run before the coverage audit of Section 6.4, narrow the gap: per-sample jitter at sensor precision (described below) and, on three equipment classes, field sensor error on temperatures, flows and positions (with 5% feedback quantisation; static pressures, humidities and powers untouched), change-of-value logging with stale repeats and gaps, and schedule shifts with holidays, each and all together, applied to the fault-free year and to every fault file and re-fitted; the false-alarm budget held in every cell (at most 3 of 96 held-out days, never more than two above the clean count), change-of-value logging was the costliest condition (one, three and no detections lost on the single-duct, dual-duct and fan coil units), and the schedule shift gained three dual-duct sensor-bias detections at no false alarm, because the one training day that set the top of the hot-deck residual band became a seeded holiday and the band narrowed, a prediction failure in the detector’s favour that is reported, not credited, and that exposes the fragility of extreme-value bands to a single day; the absence channel was not exercised.¹⁶

The field-condition arms were then repeated, pre-registered, on the shipped configurations of all five simulated systems after the coverage audit, whose 73 new residual rules include bands 0.01–0.02 units wide: the budget held in every cell (worst 7 of 96, the parallel unit under the schedule shift; no system more than two above its clean count), each condition cost at most one detection per system (the single-duct outdoor-air bias under change-of-value logging, as before; the parallel unit’s moderate airside reheat fouling under noise, change-of-value logging and their combination; the dual-duct moderate waterside cooling fouling under the combination only), nothing was gained, and the clean arm reproduced the scorecard on all five systems. The prediction that at least two of the five recoveries sitting within one band width of the healthy extremes would be lost under noise failed: one was.¹⁷ A pre-registered alternative calibration — bands at the 2nd and 98th percentiles

¹⁶outputs/x31_field_conditions.json; docs/plans/2026-09-24-x31-field-conditions-prereg.md.

¹⁷outputs/x35_field_conditions.json; docs/plans/2026-09-27-x35-field-conditions-post-coverage-prereg.md.

of the training-day scores instead of the extremes, about ten per cent narrower — gained no scenario and lost none on any system, kept all five edge recoveries, and added thirteen residual false-alarm days over 480, four each on the fan-powered units, which fired its budget falsifier; the deployed calibration is unchanged, and the edge recoveries are not artefacts of the exact extreme.¹⁸ Still, the one measured stream so far (Section 5: a field rooftop unit, 182 fault-free days) supports a false-alarm rate, 6.1 % of held-out days, and one unseen fault case, nothing more. Noise-free years make zero-false-positive thresholds cheap, bands razor-thin and silence easy; a substantial share of the project’s own recorded lessons reduce to that single observation. Two pre-registered robustness tests bound the flattery without removing it: under i.i.d. per-sample jitter at building-sensor precision (0.5 °F, 2% of flow, 0.01 of travel; not bias or drift) the re-fitted detector’s detection counts of the time are unchanged on all three systems and false alarms stay at 1–4 days in 96, and under the mirrored eight-day holdout the counts move by at most one — while the membership of the marginal scenarios shifts (at $1 \times$ jitter, two lost and two gained on the parallel unit; one and one under the mirrored holdout), which is the honest reading of a count that looks stable.¹⁹ The deployment phase on real building data remains outstanding, and until it is complete the framework should be considered validated in the laboratory rather than in the field. The contamination result of Section 7 bears directly on that gap, because a real building’s baseline year is not fault-free by construction.

Two reporting decisions require explicit disclosure. First, the deployed false-alarm rates in Table 2 treat the seasonal rate channel as corroborating evidence that may not alarm alone. The naive union of all eight channels false-alarms on 15.6% of fault-free holdout days on the single-duct system, and that channel accounts for nearly all of it. The demotion was adopted after those union figures were known, was verified to cost no detections and no change in time to detection, and both figures are published together — but it is a post-hoc choice and is identified as such. Its split-half robustness probe passes on the single-duct and parallel systems and fails on the series system, where the worst channel is the residual channel on one half and the model channel on the other, neither demoted. Readers comparing against methods that report an undemoted union should use 15.6%.

Second, until the three-way split of Section 5 the model and device channels’ holdout false-positive rows were calibration targets rather than measurements, because their thresholds were quantile-fit on the same holdout days those rows report. The split (thresholds from the eight days before each month’s holdout block; discovery on the rest) measured the size of that debt: on the two fan-powered units the model channel’s out-of-sample rows are 4 and 5 of 96 against 1 and 1 in-sample, and the deployed budgets moved from 5.2 and 4.2 to 7.3 and 6.2%; the other three systems did not move. Making the residual channel’s null a per-day rate on its own evaluable holdout days, floored at three days, then removed seven residual-only detections, one of them offset on the dual-duct unit by a new fan-law channel (the fan-powered units then evaluated 41 such days, so the floor there was 7.3 %; after the coverage audit’s added residuals they evaluate 69, floor 4.3 %): 22 of 30, 21 of 29 and 40 of 47 were the counts under the final gate before the coverage audit (26, 25 and 43 after it), and the loss fired a pre-registered falsifier and is reported as a detector change. Every reported row is now out-of-sample, though every threshold still comes from the same fault-free year.

The hybrid extension combining process-mining features with a machine-learning classifier — the first of the two out-of-scope goals named in Section 2 — has not been attempted. The deferral is deliberate rather than incidental. The adjacent literature suggests such a hybrid would raise

¹⁸outputs/x36_quantile_band.json; docs/plans/2026-09-27-x36-quantile-band-prereg.md.

¹⁹outputs/x9_x10_robustness.json; three amendments, each recorded before the result it governs; the fan-powered arms exclude the conformance channels, whose removal changes no clean count.

detection scores — Vitale and colleagues report F1 gains of thirty to seventy points from feeding alignment features to downstream classifiers in a robotics domain [51] — but its output is an anomaly score, which is precisely the artefact this framework exists to avoid. The test worth running is therefore not whether a hybrid detects more, which it very likely would, but whether it detects more at an equal false-alarm budget while still naming the implicated device. That is a phase of work with its own pre-registration, not an ablation, and it is left to future work.

The noise-floor gate applies no correction across channels and scenarios. On the order of 1,600 binomial tests are run at $\alpha = 10^{-3}$ — eight channels by 199 scenarios on six systems, with the device channel run per terminal unit where a device stratum exists — for a family-wise error near 0.8 under independence and an expectation of about 1.6 false cells across the whole study, and the per-day observations within a scenario are autocorrelated, a caveat every artefact carries. Bonferroni correction is applied only across devices within the device channel. Confidence intervals are given for the false-alarm rates and the detection counts; all are single-run figures on one simulated year per system.

One worked alarm is shown (Figure 4); it names a symptom and a location, not a root cause. The systematic scoring that exists is of the carrying rule against the seeded component and fault family (Section 6): it is subsystem-level, it names the responding component rather than the cause on the single-zone unit, and the alarm text itself has not been scored.

The benchmark families also bound what the introduction’s argument can claim. The seven datasets seed component, sensor and control-loop faults — stuck and leaking dampers and valves, sensor biases, coil fouling, airflow and refrigerant-line restriction, refrigerant charge, unstable loops and a reversed controller — and none seeds a schedule, override or setpoint-reset fault, the behaviour-based class that dominates the field survey the introduction cites. The case for sequence-aware detection on that class is therefore motivated here, not tested; the detections reported are of faults a rule library could in principle enumerate, and Section 9.1 measures how many of them one such library does.

Beyond these, the long detection tail is a real weakness for gradually developing faults; the comparison against deployed rule-based practice covers three systems and one third-party rule library (Section 9.1); and six simulated systems across five equipment classes demonstrate configuration-level portability without establishing generality across the diversity of commercial HVAC equipment. A collection of eight further labelled AHU fault datasets published in 2026 [16] is the obvious next external-validity target.

9.1 The comparison against commercial practice

The comparison that matters most to practitioners is against the rule-based detection actually deployed in commercial building automation systems [20]. Its result, stated first: held to the treatment STRATA gives its own signature rules, a correctly wired third-party Guideline 36 battery detects 12 of 14 on the single-duct air handler at one false-alarm day against STRATA’s 13 at one, and 8 of 30 and 9 of 29 on the fan-powered terminal units at no false-alarm day against STRATA’s 26 and 25 at eight and nine. STRATA is within one detection and one false-alarm day of current practice on the air handler and detects three times as many seeded faults on the terminal units, paying eight or nine false-alarm days in 96 for them; the difference on the terminal units is coverage — the per-zone residual readers of Section 6.4 — rather than the gate. The pre-registered refutation condition, that the battery matching STRATA’s count at an equal or lower false-alarm rate on any system would end the improvement claim there, fired on every system, and the claim is withdrawn. The rest of this section says how the comparison was built, what voided its first two executions, and what a reader must weigh. The design imports a third-party open-source implementation of the

Guideline 36 fault conditions, open-fdd 4.4.1 [35], rather than reimplementing them, since writing one’s own version of a competing method invites the objection that the comparison was won by construction; parameters stay at the library’s installed defaults; air handlers are graded with the standard’s fifteen air-handler conditions and terminal units with the library’s terminal-unit rules, run per zone; and both methods sit the identical examination.

Two executions preceded the one reported. The first (2026-09-11) fed two coil conditions water-side temperatures where the standard specifies air-side measurements, graded terminal-unit faults with the air-handler rulebook, and fed the adapter the static-pressure columns documented as provenance-leaking in Section 8; it was voided. The second, under a pre-registered repair, held every prediction — the battery flagged 77, 96 and 96 of 96 fault-free holdout days and detected nothing under the gate — and was then voided by two independent audits that found the repair’s own defects: a fan-speed column that is a dead constant in the dataset (which the project’s sensor map already recorded), a terminal-unit gate on the box fan of a parallel unit (which runs only while heating), a hunting rule reading a binary fan status as a hunting loop, night-cycle graded as occupied, a single-duct leaving-air rule applied to boxes that mix plenum air by design, a minimum-outdoor-air condition reading a column not recorded in cfm, and, in the pre-registered design itself, a gate applied once to the pooled union of all rules, which is a tautology once the union flags every healthy day. The amendment recording all of this was committed before the third execution, whose artefacts are the ones quoted.²⁰

Three framings were pre-registered in the amendment and the falsifier applied to each. Gated once on the pooled union, as first written, the battery detects 1, 0 and 0 scenarios, because its union false-alarms on 49, 96 and 96 of 96 holdout days. Gated per rule against each rule’s own holdout false alarms — STRATA’s per-channel structure — it detects 12 of 14 on the single-duct unit and every scenario on both fan-powered units; but on those units the three air-handler conditions that carry the verdict flag the identical day sets on the fault-free file and on every fault file, and pass only because each rule’s holdout rate sits below its full-year rate, a non-stationarity the holdout floor cannot see and that STRATA’s channels never present because onboarding silences every rule on the whole fault-free year first. That framing is reported as a property of the floor, not as detection; the lesson it records is that a holdout null is valid only for a channel that is stationary across the year, which is what silence on the fault-free year buys. The third framing treats the battery exactly as STRATA treats its own signature rules: keep the rules that are silent on the fault-free year and hold them to the same 3/365 null. It gives the counts stated at the head of this section: 12 of 14 on the single-duct air handler at one false-alarm day (the two misses are the 10% and 25% stuck dampers, seen only by a condition that is not silent), and 8 of 30 and 9 of 29 on the terminal units at no false-alarm day, the battery’s detections being the unstable room loop, the larger stuck and leaking reheat valves and the stuck dampers.

A re-audit of that execution found no adapter defect but three things a reader must weigh: the per-rule framing admits rules that detect the fault-free file itself, and without them the fan-powered counts are 15 of 30 and 17 of 29; the silent-rule selection is made on the whole fault-free year including the holdout, so the battery’s one false-alarm day is a post-selection count, whereas STRATA’s one is a frequency-channel day against a band fitted on training days only — though STRATA’s own signature rules are silenced on the same whole year, so the rules-channel zero is post-selection on both sides; and the single-duct outdoor-air-bias file, adjudicated away on STRATA’s side (Section 8), is counted on the battery’s, where its carrying condition passes the same branch-immunity control — under every reading STRATA detects more on that unit, and the match rests

²⁰docs/plans/2026-09-27-x38-guideline36-repaired-prereg.md and its Amendment 1; outputs/x38_guideline36.json; earlier executions kept under outputs/void/ and as *_run1.json.

on the false-alarm tie alone. The falsifier fired on every system: on the air handler because equal false alarms is a match, on the terminal units because none is fewer. The pre-registered claim cannot be made; what the artefact supports is the narrower statement at the head of this section. Three caveats travel with the numbers. The parameters are open-fdd’s installed defaults, which are tighter on every condition that fired than the Guideline 36 values the library’s own parameter labels cite (sensor errors 1.15 °F against 2–5, fan heat rise 0.55 against 2 °F, no mode-change delay against 30 minutes); with the standard’s values the fan-powered air handler’s 2.15 °F fan heat rise would largely not trip the mixing-envelope condition. The terminal rules carry fixed comfort bands and flow thresholds that do not read the building’s own setpoint columns. And supplying the one site datum every deployment supplies (the design minimum outdoor-air flow, taken from the fault-free year) left the outdoor-air-fraction condition alarming on 94 of 96 healthy days, because the implementation carries no operating-state gate.

10 Conclusion

STRATA works on the benchmarks it was tested on. A stratified event log built from the control sequence, process models discovered at device and unit strata, and eight calibrated, physically interpretable channels detect 158 of 175 seeded faults on five public benchmark systems (157 once one dataset defect is adjudicated) at a median of one day, inside a stated false-alarm budget of 26 days in 480 that holds under injected field conditions on every system and on one measured rooftop unit; two of the five systems were onboarded blind by configuration alone; each alarm names the correct terminal unit in 37 of 50 checkable detections and a wrong one in none, names the faulted subsystem in 83 % of all detections, and carries the evidence behind it — properties a scalar anomaly score of equal detecting power cannot supply. Against current practice, a correctly wired Guideline 36 battery under the same gate is within one detection and one false-alarm day on the air handler, and STRATA detects three times as many faults on the terminal units at eight or nine false-alarm days; the pre-registered improvement claim fired its falsifier and is stated as that.

The process-mining question that motivated the work is settled, and the answer is the division of labour the introduction states. Process mining had not previously been applied to fault detection in building HVAC systems, an absence that the one systematic review whose scope could have captured such work also reports; this work made the application and measured what it contributes. Process mining helps to get detection but is not detecting on its own: the strata and the event abstraction carry the localisation, invariants discovery surfaced became several of the rules that detect, while conformance checking of the discovered nets did not out-detect hand-written rules, removing it from the deployed detector changes no result, and the one test that might have given the nets a predictive role was found, by its own amendment, to have no power and is reported as such.

The contribution this paper is surest of is the protocol that made both claims checkable: pre-registration with binding falsifiers, independent recomputation before any number is quoted, and automated tests that pin not only published values but fired refutations and disclosure obligations. Its evidence is what it did to the project’s own claims. Four pre-registered hypotheses were refuted, three results withdrawn, one pre-registered test found to have no power and its powered re-run matched by a discovery-free control, and one fault-informed rule whose falsifier fired, and all ten are reported, as are the four later falsifiers that bounded component attribution, the diagnosis layer, quantile bands and the comparison against current practice; the numbered lessons are recorded with the standing rule each produced; eight defects in a widely used public benchmark were documented with reproducible evidence; and an adversarial audit voided a completed comparison against com-

mercial practice rather than allowing it to be published, and the repaired comparison then fired its own falsifier and is reported as such. In a subfield where independent verification is rare, a result that has survived a genuine attempt at refutation is worth more than a larger number that has not.

Acknowledgements

The author thanks Dr. Anthony Aighobahi and Dr. Mridula Sharma for supervision throughout this project. AI coding assistants were used for implementation, analysis scripting, and adversarial auditing of results; all experimental designs, pre-registrations, and claims were reviewed and approved by the author.

A Every pre-registered experiment

Tables 9 and 10 list every pre-registered experiment in the order it was run, with its pre-registered prediction, its outcome and what fired. Each row resolves to a ledger under outputs/ or, where noted in the repository’s research log, to the phase report; amendments are folded into the row they amend.

References

- [1] Mohammad Amin Fatemi Abdollah, Rossano Scoccia, and Marcello Aprile. Transformer encoder based self-supervised learning for HVAC fault detection with unlabeled data. *Building and Environment*, 258:111568, 2024.
- [2] Katsiaryna Akhramovich, Estefanía Serral, and Carlos Cetina. A systematic literature review on the application of process mining to Industry 4.0. *Knowledge and Information Systems*, 66(5):2699–2746, 2024.
- [3] Alessandro Berti, Sebastiaan van Zelst, and Daniel Schuster. PM4Py: A process mining library for Python. *Software Impacts*, 17:100556, 2023.
- [4] Fábio Bezerra and Jacques Wainer. Algorithms for anomaly detection of traces in logs of process aware information systems. *Information Systems*, 38(1):33–44, 2013.
- [5] Jian Bi, Hua Wang, Enbo Yan, Chuan Wang, Ke Yan, Liangliang Jiang, and Bin Yang. AI in HVAC fault detection and diagnosis: A systematic review. *Energy Reviews*, 3(2):100071, 2024.
- [6] Edyta Brzychczy, Milda Aleknonytė-Resch, Dominik Janssen, and Agnes Koschmider. Process mining on sensor data: a review of related works. *Knowledge and Information Systems*, 67:4915–4948, 2025.
- [7] Joos C. A. M. Buijs, Boudewijn F. van Dongen, and Wil M. P. van der Aalst. On the role of fitness, precision, generalization and simplicity in process discovery. In *On the Move to Meaningful Internet Systems: OTM 2012 (CoopIS)*, volume 7565 of *Lecture Notes in Computer Science*, pages 305–322. Springer, 2012.
- [8] Josep Carmona, Boudewijn van Dongen, Andreas Solti, and Matthias Weidlich. *Conformance Checking: Relating Processes and Models*. Springer, 2018.

- [9] Armando Casillas, Yimin Chen, Jessica Granderson, Guanjing Lin, Zhelun Chen, Jin Wen, and Sen Huang. Development of high-fidelity air handling unit fault models for FDD innovation: lessons learned and recommendations. *Journal of Building Performance Simulation*, 17(5):615–630, 2024.
- [10] Eliot Crowe, Yimin Chen, Hayden Reeve, David Yuill, Amir Ebrahimifakhar, Yuxuan Chen, Lucas Troup, Amanda Smith, and Jessica Granderson. Empirical analysis of the prevalence of HVAC faults in commercial buildings. *Science and Technology for the Built Environment*, 29(10):1042–1057, 2023.
- [11] Wael Deabes, Kheir Eddine Bouazza, and Wasl Alghthami. Smart fuzzy Petri net-based temperature control framework for reducing building energy consumption. *Sensors*, 23(13):5985, 2023.
- [12] Qiao Deng, Zhiwen Chen, Wanting Zhu, Zefan Li, Yifeng Yuan, and Yalin Wang. Adaptive fusion graph convolutional network based interpretable fault diagnosis method for HVAC systems enhanced by unlabeled data. *Energy and Buildings*, 324:114901, 2024.
- [13] Karim El Mokhtari and J. J. McArthur. Autoencoder-based fault detection using building automation system data. *Advanced Engineering Informatics*, 62:102810, 2024.
- [14] Stephen Frank, Guanjing Lin, Xin Jin, Rupam Singla, Amanda Farthing, and Jessica Granderson. A performance evaluation framework for building fault detection and diagnosis algorithms. *Energy and Buildings*, 192:84–92, 2019.
- [15] Naghmeh Ghalamsiah, Jin Wen, K. Selcuk Candan, Teresa Wu, Zheng O’Neill, and Asra Aghaei. Unsupervised domain adaptation for HVAC fault diagnosis using contrastive adaptation network. *Energy and Buildings*, 337:115659, 2025.
- [16] Naghmeh Ghalamsiah, Jin Wen, Guanjing Li, Yimin Chen, Xing Lu, Yangyang Fu, et al. Labeled datasets for air handling units operating in faulted and fault-free states. *Scientific Data*, 13(15), 2026.
- [17] Luís Gomes, Anikó Costa, João Paulo Barros, and Paulo Lima. Petri net based building automation and monitoring system. In *5th IEEE International Conference on Industrial Informatics (INDIN)*, 2007.
- [18] Jessica Granderson, Guanjing Lin, Yimin Chen, Armando Casillas, Jin Wen, Zhelun Chen, Piljae Im, Sen Huang, and Jiazhen Ling. A labeled dataset for building HVAC systems operating in faulted and fault-free states. *Scientific Data*, 10:342, 2023.
- [19] Jessica Granderson, Guanjing Lin, Ari Harding, Piljae Im, and Yan Chen. Building fault detection data to aid diagnostic algorithm creation and performance testing. *Scientific Data*, 7:65, 2020.
- [20] Jessica Granderson, Rupam Singla, Ebony Mayhorn, Paul Ehrlich, Draguna Vrabie, and Stephen Frank. Characterization and survey of automated fault detection and diagnostic tools. Technical Report LBNL-2001075, Lawrence Berkeley National Laboratory, 2017.
- [21] H. Burak Gunay, Jayson Bursill, Brent Huchuk, and Scott Shillinglaw. Inverse model-based detection of programming logic faults in multiple zone VAV AHU systems. *Building and Environment*, 211:108732, 2022.

- [22] Jake M. Hofman, Angelos Chatzimparmpas, Amit Sharma, Duncan J. Watts, and Jessica Hullman. Pre-registration for predictive modeling, 2023. arXiv:2311.18807.
- [23] Erik Johannes Husom, Arda Goknil, Felix Mannhardt, Simeon Tverdal, Sagar Sen, and Phu Nguyen. Unsupervised learning and process analysis for sensor data validation in the IIoT. *ACM Transactions on Internet Technology*, 26(2):Article 17, 2026.
- [24] Dominik Janssen, Felix Mannhardt, Agnes Koschmider, and Sebastiaan J. van Zelst. Process model discovery from sensor event data. In *Process Mining Workshops (ICPM 2020)*, volume 406 of LNBIP, pages 69–81. Springer, 2021.
- [25] Ivo Jimenez, Michael Sevilla, Noah Watkins, Carlos Maltzahn, Jay Lofstead, Kathryn Mohror, Andrea Arpaci-Dusseau, and Remzi Arpaci-Dusseau. The Popper convention: Making reproducible systems evaluation practical. In *IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW)*, 2017.
- [26] Sander J. J. Leemans, Dirk Fahland, and Wil M. P. van der Aalst. Discovering block-structured process models from event logs — a constructive approach. In *Application and Theory of Petri Nets and Concurrency (PETRI NETS 2013)*, volume 7927 of *Lecture Notes in Computer Science*, pages 311–329. Springer, 2013.
- [27] Sander J. J. Leemans, Dirk Fahland, and Wil M. P. van der Aalst. Discovering block-structured process models from event logs containing infrequent behaviour. In *Business Process Management Workshops (BPM 2013)*, volume 171 of *Lecture Notes in Business Information Processing*, pages 66–78. Springer, 2014.
- [28] A. Mahmoudan, P. Nojedehe, and B. Gunay. Fault detection and severity classification in HVAC fan coil units using a hybrid autoencoder-based method. *Journal of Physics: Conference Series*, 3140(2):022028, 2025.
- [29] Araham Jesus Martinez Lagunas and Mazdak Nik-Bakht. Process mining, modeling, and management in construction: A critical review of three decades of research coupled with a current industry perspective. *Journal of Construction Engineering and Management*, 150(11), 2024.
- [30] Azin Moradbeikie, Ana Paula Ayub da Costa Barbon, Iuliana Malina Grigore, Douglas Fernandes Barbin, and Sylvio Barbon Junior. Process mining of sensor data for predictive process monitoring: A HACCP-guided pasteurization study case. *Systems*, 13(11):935, 2025.
- [31] Azin Moradbeikie, Iuliana Malina Grigore, Sergio Ivan Lopes, and Sylvio Barbon Junior. Sensor2EventLog: Bridging continuous IoT data and process mining through eventization. In *Advanced Information Systems Engineering, Lecture Notes in Computer Science*. Springer, 2026.
- [32] Adil Mukhtar, Michael Hadwiger, Franz Wotawa, and Gerald Schweiger. Reproducibility of machine learning-based fault detection and diagnosis for HVAC systems in buildings: An empirical study, 2025. arXiv preprint 2508.00880; finds two of sixty-five studies shared code links, one of them broken.
- [33] Jorge Muñoz-Gama and Josep Carmona. A fresh look at precision in process conformance. In *Business Process Management (BPM 2010)*, volume 6336 of *Lecture Notes in Computer Science*, pages 211–226. Springer, 2010.

- [34] Timo Nolle, Stefan Luetzgen, Alexander Seeliger, and Max Mühlhäuser. BINet: Multi-perspective business process anomaly classification. *Information Systems*, 103:101458, 2022.
- [35] open-fdd contributors. open-fdd: open-source fault detection and diagnostics for HVAC implementing the ASHRAE Guideline 36 fault conditions, 2025. PyPI package; the Guideline 36 battery imported, not reimplemented, for the comparison in this paper.
- [36] Samuel Pawel, Lucas Kook, and Kelly Reeve. Pitfalls and potentials in simulation studies: questionable research practices in comparative simulation studies allow for spurious claims of superiority of any method, 2022. arXiv:2203.13076.
- [37] Cameron Pitsch, Tobias Brockhoff, Jan Niklas Adams, Sander J. J. Leemans, Leo Anthony Celi, and Wil M. P. van der Aalst. Hypothesis testing for processes. In *2025 7th International Conference on Process Mining (ICPM)*, pages 1–8. IEEE, 2025.
- [38] Ojas Pradhan, Jin Wen, Yimin Chen, Xing Lu, Mengyuan Chu, Yangyang Fu, Zheng O’Neill, Teresa Wu, and K. Selçuk Candan. Dynamic bayesian network-based fault diagnosis for ASHRAE guideline 36. In *Proceedings of the 8th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation (BuildSys ’21)*, pages 365–368. ACM, 2021.
- [39] Prince, Byungun Yoon, and Prashant Kumar. Fault detection and diagnosis in air-handling unit (AHU) using improved hybrid 1D convolutional neural network. *Systems*, 13(5):330, 2025.
- [40] Marco Pritoni, Guanqing Lin, Yimin Chen, Raphael Vitti, Christopher Weyandt, and Jessica Granderson. From fault-detection to automated fault correction: A field study. *Building and Environment*, 214:108900, 2022.
- [41] Anne Rozinat and Wil M. P. van der Aalst. Conformance checking of processes based on monitoring real behavior. *Information Systems*, 33(1):64–95, 2008.
- [42] Norkobil Saydirasulovich Saydirasulov, Abror Shavkatovich Buriboev, Shuxrat Isroilov, Ryumduck Oh, Shavkat Buribayev, Abbos Abduvaytov, Jamshid Umirov, Jasur Ismailovich Badalov, Aziza Axmedova, Cheolwon Lee, and Heung Seok Jeon. A reproducible benchmark-validity audit and calibration study for cross-home fault diagnosis in smart-home sensor systems. *Sensors*, 26(16):5025, 2026.
- [43] Niek Tax, Xixi Lu, Natalia Sidorova, Dirk Fahland, and Wil M. P. van der Aalst. The imprecisions of precision measures in process mining. *Information Processing Letters*, 135:1–8, 2018.
- [44] Wunna Tun, Johnny Kwok-Wai Wong, and Sai-Ho Ling. Hybrid random forest and support vector machine modeling for HVAC fault detection and diagnosis. *Sensors*, 21(24):8163, 2021.
- [45] Wil M. P. van der Aalst. *Process Mining: Data Science in Action*. Springer, 2nd edition, 2016.
- [46] Wil M. P. van der Aalst, Arya Adriansyah, and Boudewijn van Dongen. Replaying history on process models for conformance checking and performance analysis. *WIREs Data Mining and Knowledge Discovery*, 2(2):182–192, 2012.
- [47] Sebastiaan J. van Zelst, Felix Mannhardt, Massimiliano de Leoni, and Agnes Koschmider. Event abstraction in process mining: literature review and taxonomy. *Granular Computing*, 6(3):719–736, 2021.

- [48] Seppe K. L. M. vanden Broucke, Jochen De Weerd, Jan Vanthienen, and Bart Baesens. Determining process model precision and generalization with weighted artificial negative events. *IEEE Transactions on Knowledge and Data Engineering*, 26(8):1877–1889, 2014.
- [49] Emilia Villani and Paulo E. Miyagi. A hybrid Petri net modeling approach for HVAC systems in intelligent buildings. *SBA: Controle & Automação*, 15(2), 2004.
- [50] Francesco Vitale, Nicola Dall’Ora, Sebastiano Gaiardelli, Enrico Fraccaroli, Nicola Mazzocca, and Franco Fummi. Process mining-driven modeling and simulation to enhance fault diagnosis in cyber-physical systems. *Journal of Manufacturing Systems*, 2025. arXiv:2506.21502.
- [51] Francesco Vitale, Marco Pegoraro, Wil M. P. van der Aalst, and Nicola Mazzocca. Control-flow anomaly detection by process mining-based feature extraction and dimensionality reduction. *Knowledge-Based Systems*, 310:112970, 2025.
- [52] Hong Wang, Wenjing Yan, Yajuan Zhang, Jianqiang Yu, Wensheng Gao, Bowen Zhou, and Yi Liu. xMHAC-PDCNN: An explainable deep network with spatial feature embedding for fault detection and diagnosis of chillers under variable operating conditions. *Building and Environment*, 284:113504, 2025.
- [53] Hans Weytjens and Jochen De Weerd. Creating unbiased public benchmark datasets with data leakage prevention for predictive process monitoring. In *Business Process Management Workshops (BPM 2021)*, volume 436 of *Lecture Notes in Business Information Processing*, pages 18–29. Springer, 2022.
- [54] Xiaodong Xu et al. Wireless sensors system for the chillers fault diagnostic based on characteristic Petri net modelling. *International Journal of Wireless and Mobile Computing*, 9(3), 2015.
- [55] David P. Yuill and James E. Braun. Evaluating the performance of fault detection and diagnostics protocols applied to air-cooled unitary air-conditioning equipment. *HVAC&R Research*, 19(7):882–891, 2013.
- [56] Yinzhen Zhong and Alexei Lisitsa. Can process mining help in anomaly-based intrusion detection? arXiv:2206.10379, 2022. Preprint; not peer reviewed. Reports that naïve process mining on network event logs was ineffective for intrusion detection.
- [57] Xu Zhu, Kang Chen, Burak Anduv, Xinqiao Jin, and Zhimin Du. Transfer learning based methodology for migration and application of fault detection and diagnosis between building chillers for improving energy efficiency. *Building and Environment*, 200:107957, 2021.

Table 1: The eight channels as deployed. Every threshold is set from the fault-free year alone: bands from its training days, conformance thresholds from its calibration slice, and the noise floor from its held-out days. “Alphabet” names which side of the wall the channel reads. The rate channel is advisory: reported, never counted in the deployed union.

Channel	Alphabet	A day is flagged when	Threshold source	In union
Rules	signature	a control-sequence rule (Guideline 36 / APAR lineage) fires during operation	fixed bands from the sequence documentation; no fitted parameter	yes
Residual	raw signals	a per-mode physical residual (coil-off supply-mixed rise, outdoor-air fraction, static $\div$ speed ² , refrigerant approach) leaves its healthy band	[min, max] of training-day scores, widened to at least the 0.5 °F sensor floor; exceedance must also clear that floor	yes
Model conformance	state	the day’s trace aligns to the unit net with fitness below threshold	1 % quantile of per-day fitness on the calibration slice	yes
Device conformance	state	an actuator’s daily lifecycle trace aligns to its device net below threshold	as above, per device net	yes
Absence	state	a device that is active on scheduled training days produces no lifecycle event on a scheduled day	each device’s healthy silence rate on training days	yes
Frequency	state	a daily activity count (unit or device) leaves its healthy range	[min, max] over training days, for activities seen on ≥ 5 % of them	yes
Oscillation	raw signals	an actuator’s daily count of direction reversals leaves its healthy range	[min, max] over training days	yes
Rate	state	a 30-day active-day rate leaves the range of healthy windows ending in the same calendar month	training-year windows, month-matched	advisory

	SDAHU	PFPU	SFPU
Scenarios detected	14/14 naive; 13/14 adjudicated	26 / 30	25 / 29
Median time to detection	1 day	1 day	1 day
Deployed false-alarm rate	1.0%	8.3%	9.4%
Naive all-channel union	15.6%	15.6%	9.4%
Wrong alarms / all alarms	0.03%	0.11%	0.12%

Table 2: Detection performance over the 96-day healthy holdout (the last eight days of each month). Naive and adjudicated refer to before and after the branch adjudication of Section 7; the cry-wolf ratio is wrong alarms as a share of all alarms ($\leq 0.13\%$ on every system, from 96 fault-free holdout days against 4,540–9,490 observed fault-scenario days), and is small partly because observed fault-scenario days outnumber fault-free days 47–99 to 1. The SDAHU score is reported in both forms for the reason given in Section 7; the SFPU universe excludes one file with a rotated calendar (erratum E4); the two false-alarm rows differ by the demotion disclosed in Section 9 on the single-duct and parallel systems and coincide on the series unit, where the demoted channel contributes no false-alarm day. Since the three-way split every deployed rate is out-of-sample (Section 5).

System (LBNL)	points	rules (state / sig.)	scored	detected	false alarms	healthy days
SDAHU, single-duct AHU	26	3 / 12	14	14 (13 adj.)	1 / 96	365
PFPU, parallel fan-powered box	109	13 / 45	30	26	8 / 96	365
SFPU, series fan-powered box	109	13 / 49	29	25	9 / 96	365
DDAHU, dual-duct AHU	114	15 / 56	55	50	3 / 96	365
FCU, fan coil unit	29	7 / 19	47	43	5 / 96	365
RTU, simulated rooftop unit	23 [†]	3 / 5	24	20	0 / 28	100
RTU, measured rooftop unit (Site 2)	26 [†]	2 / 7	1 case	—	3 / 49	182

Table 3: The seven LBNL datasets under the final gate: what each configuration maps and declares after the coverage audit of Section 6.4, what was scored, and the out-of-sample false-alarm count on the held-out fault-free days (the last eight of each month). The measured unit carries one fault case and is used for its false-alarm rate only. [†]Includes one derived operating-state column (fan-on for the measured unit; a constant for the simulated unit, which has no schedule).

System	Detected	At 1 day	At 2 days	Remaining TTDs (days)
SDAHU	13 (adjudicated)	9	4	0
PFPU	26	26	0	—
SFPU	25	25	0	—

Table 4: Time-to-detection distribution behind the one-day median, by system.

	STRATA	PCA-SPE (strict)
Detected (of 73, adjudicated)	64	61
Per system	13 / 26 / 25	13 / 23 / 25
Holdout false-alarm rate	1.0% [0.2, 5.7] / 8.3% [4.3, 15.6] / 9.4% [5.0, 16.9]	1.3% [0.2, 6.8] / 1.0% [0.2, 5.7] / 1.0% [0.2, 5.7]
Sensor columns mapped (STRATA)	26 of 30 and 109 of 109 (12 and 56 before the coverage audit)	—
Raw daily features consumed (PCA)	—	108 and 424
Implicated device named correctly	37 of 50 (13 unnamed, 0 wrong, 1 indeterminate; the 14 single-duct detections have no device stratum; naive universe)	not applicable
Names the implicated device	yes	no

Table 5: STRATA against the principal-component baseline. Per-system entries are ordered SDAHU / PFPU / SFPU; the counts behind the false-alarm rates are 1/96, 8/96 and 9/96 for STRATA and 1/79, 1/96 and 1/96 for the baseline, with Wilson 95% intervals in brackets. Feature counts are given for the single-duct and the fan-powered systems.

System	columns mapped	detected, before → after	false alarms, before → after	lost
SDAHU	12 → 26 of 30	14 → 14 / 14	1 → 1	0
PFPU	56 → 109 of 109	22 → 26 / 30	7 → 8	0
SFPU	56 → 109 of 109	21 → 25 / 29	6 → 9	0
DDAHU	79 → 114 of 114	45 → 50 / 55	3 → 3	0
FCU	29 → 29 of 29	40 → 43 / 47	4 → 5	0
all	232 → 387 of 391	142 → 158 / 175	21 → 26 of 480	0

Table 6: The coverage audit. Sixteen scenarios recovered, none lost, five false-alarm days added over 480 held-out days, the model and device rows identical on every system (the log hashes change, because the new readers emit signature events). Totals use the naive single-duct count. Ledgers outputs/x33_coverage_*.json; per-rule credits outputs/x33_residual_credits_*.json.

#	Hypothesis or result	Refutation condition / what fired	What survived
1	Hypothesis: discovered models out-detect hand-written rules.	A one-line matched rule — occupied workday and AHU heating never active — covered 192 days against the model channel’s 135 when the falsifier fired (178 in the current artefact, after the threshold repairs) on the same scenario, with zero false alarms on 365 healthy days. The falsifier fired.	The models were design-time scaffolding: an invariant surfaced during discovery, the expert wrote the rule, and thereafter the models contributed no detection. The matched rule is a permanent ablation arm.
2	Hypothesis: behavioural structure discovered on one building supports detection on another, zero-shot.	The detection hypothesis failed its pre-registered falsifier, yielding zero to one detection per rotation under the confidence-bound gate (the point-estimate gate gives 0, 1 and 5).	Real shared structure: fan-powered-unit logs outscore their own shuffled controls on the single-duct net by 0.15 and 0.18 in mean fitness. Portability is claimed at configuration level only.
3	Result: the conformance channels detect faults no other channel catches.	Repeated audits found the discovered nets order-permissive on loop-heavy alphabets, and every genuine model-channel detection was a frequency deviation surfacing through alignment cost.	The frequency channel, which measures those deviations directly and two-sidedly; the conformance channels had nothing left that was uniquely theirs.
4	Result: the single-duct system scores 14 of 14.	An audit established that the dataset’s fault-free file was simulated on a different configuration branch than every fault file; under a branch-corrected band, one scenario’s only deployed channel produced zero flags on its 148 evaluable window days.	13/14 adjudicated, quoted together with 14/14 naive, and the standing rule that healthy-versus-fault divergence is fault evidence only after a fault-versus-fault control.
5	Result: a day-level significance test adopted in Phase 4.	The test rested on an impossible null — the discordant cell it tested was zero by construction — so it awarded significance to noise.	A binomial against each channel’s own measured holdout rate, after thirteen scenario rows on the parallel unit lost their stars. The standing rule is that the null is always the channel’s own noise.
6	Hypothesis: healthy silence tuned at one-minute sampling is preserved at fifteen-minute sampling.	Configurations tuned at one-minute sampling gain 28 to 64 spurious healthy signature-days at fifteen-minute sampling and one residual coverage drops by more than a quarter; four falsifiers fired. A separate pre-registered measurement of a known property is not counted here because it confirmed its hypothesis: a single score-carrying contaminated day collapses the two negative rungs of the coil-bias ladder from 164 detections out of 164 to none, by dragging the band edge to -7.19 .	Contamination from a mild source announces itself on 100% of the contaminated days (5/5, 13/13 and 27/27 signature days), whereas the dangerous source is visible on none of its 27. The transfer claim was scoped so that the healthy-silence gate is re-run at the deployment sampling rate.
7	Protocol failure: a pre-registered test of whether discovery predicts where a rule transfers.	The model quantity is constant within a building, so nine cells were three buildings pseudo-replicated; the valid null over three buildings has minimum $p = 0.167$ against $p < 0.05$. Never had power; the seal covered	No conclusion in either direction. The lesson: pre-registration did not fix the unit of analysis.

ID	System	Columns / files	Defect	Consequence for evaluation
E1	SDAHU	oa_bias (four severity files)	One simulation run shipped four times under different severity labels; the labelled sensor-side bias never appears in the logged column. The evidence is consistent with a bias injected only into the cooling interlock’s outdoor-air input, but observationally indistinguishable from a mislabelled cooling-lockout-setpoint fault.	The four severity labels index one run, not four independent scenarios, and the labelled sensor-side bias is absent from the logged column; the mechanism is not asserted.
E2	SDAHU	coi_leakage (four severity files)	One simulation run shipped four times under different severity labels (the scored count of 14 already collapses each such family to one scenario).	The four severity labels index one run, not four independent scenarios.
E3	SDAHU	SA_SP, SA_SPSPT	A constant of about 401.86 added to SA_SP in the fault-free file and subtracted from SA_SPSPT in every fault file — one column corrupted per branch, a different column each — with no overlap between the two ranges.	Either column alone separates healthy from fault data by file provenance rather than by fault physics; both are excluded from all baselines here.
E4	SFPU	SensorBias_ RMTEMP_-2C	The file’s calendar is rotated relative to its siblings.	Excluded here from family claims, so the SFPU scenario universe is scored without it.
E5	SDAHU	healthy file against every fault file	The healthy file was simulated on a different configuration branch than every fault file, with an occupied damper floor of 0.0 against 0.1 and different schedules.	Healthy-versus-fault divergence partly measures the branch rather than the fault; one single-duct scenario was detected through that provenance difference (14/14 naive, 13/14 adjudicated).
E6	FCU	Fouling_ Cooling_ Airside_Minor, Fouling_ Heating_ Airside_Minor	The two files are byte-identical: one simulation run shipped under two different family labels (a cooling coil and a heating coil).	One label is wrong and the data cannot say which; the content is scored once and the second entry is excluded, so the fan coil unit’s fouling family is 11 scenarios and its scored total 47.
E7	PFPU, every fault file	SFPU	The documentation places every seeded fault in the west zone (_W); in every file only the south zone’s columns (_S) move (57 of 60 files by the movement score, three indeterminate; the project’s earlier zone audit, a different instrument, gives 50 and 10).	A reader scoring localisation against the documented zone scores every correct alarm as wrong; the project’s zone ground truth was established from the data.
E8	DDAHU	SensorBias_ CSP/HSP_ -2inwg, -4inwg	The deck static column is unchanged; the four mixing-box statics shift by +0.20 and +0.40 in.wg, one tenth of the labelled bias (the +2 °C temperature file shifts by exactly its label).	Severity ladders and per-in.wg sensitivity claims built from these labels are off by ten; detections are unaffected.

Table 8: Eight machine-verified defects in the LBNL benchmark datasets. Each has a reproducing command in the repository’s ERRATA.md.

ID	Question	Pre-registered prediction	Outcome	Fired / failed
—	Do discovered unit models out-detect a hand-written rule?	The model channel covers more fault days than any matched rule	A one-line matched rule covered 192 days against the model channel’s 135 (178 in the current artefact), at 0 of 365 healthy false alarms	Falsifier fired (matched-rule ablation)
—	Does behavioural structure transfer zero-shot to a new building?	Source-fitted bands detect faults on the target system	0 of 14 single-duct scenarios detected under the confidence-bound gate	Falsifier fired (cold-start detection)
—	Does model support predict where a transplanted rule holds? Three amendments.	Spearman $\rho < 0$ with permutation $p < 0.05$ over nine cells	$\rho = -0.69$, $p = 0.200$; the valid null over three buildings has minimum $p = 0.167$	F1 and F2 fired, then both voided; never had power
X5	Are severity ladders monotone in dose?	No ladder is significantly anti-monotone	Series-unit fouling $\rho = +0.87$; no anti-monotone ladder; F-X5.a nearly unfireable at $n = 5$	None
X7	Does healthy silence survive 15-minute resampling?	Healthy signature days and residual coverage are preserved	Healthy years gain 28, 64 and 64 signature days; one residual’s coverage falls 31 to 1	F-X7.a (three systems) and F-X7.b
X8	Does training contamination break extreme-value calibration?	Breakdown at small contamination; mild sources stay visible to rules	One score-carrying contaminated day collapses both negative coil-bias rungs, 164 of 164 to 0	None
X9/ X10	Is the deployed detector robust to sensor jitter and to the holdout split?	Detections change by at most 3; false alarms under 10 %	Counts unchanged at $1 \times$ jitter, within one on the mirrored split; 0 to 4 false alarms of 96	None
X11	Is the single-duct OA-bias detection the fault or the configuration branch?	Five baseline estimators agree; the branch offset explains the flags	Offset $+1.06^\circ\text{F}$; OA-bias drops to 0 flags on 148 window days; 14/14 becomes 13/14	F-X11.d fired on run 1; leg re-specified
X12	Does the model’s time perspective recover missed scenarios?	The duration channel newly detects missed scenarios	11 scenarios significant but 0 newly detected; unique against frequency on 11	None; P3 failed without one
X13	Does a water-side coil-effectiveness residual separate reheat fouling?	At least 6 of the 9 fouling scenarios it counted significant	1 of the 12 fouling files scored significant, already detected elsewhere; nothing newly detected	F-X13.b
X14	Was the discovered model starved of vocabulary?	At least one newly detected scenario inside the false-alarm budget	One newly detected; model channel fires on 14 scenarios against 5; 7, 18 and 21 false alarms of 96	F-X14.a (budget)
X15	Does the enriched gain survive frequency-only scoring under two splits?	The gain is significant under both splits and inside budget	The same one scenario is significant under both splits; 41 significant against 12 deployed	None
X17	Can a dual-duct air handler be onboarded by configuration alone?	Config-only, gates clean, at least 60 % of scenarios detected	45 of 55 detected (81.8 %) at 3 false alarms of 96; one silence iteration	None; P6 failed (5 fouling, not ≤ 4)

Table 9: Every pre-registered experiment, its prediction, its outcome and what fired (part 1 of 2).

ID	Question	Pre-registered prediction	Outcome	Fired / failed
X18	Does the enriched-frequency gain replicate on the dual-duct unit?	A missed fouling scenario becomes significant under both splits	0 newly significant; enriched 22 against deployed 8 on already-detected scenarios	F-X18.b
X19 run 1	Can a fan coil unit be onboarded by configuration alone?	Gates clean, at most 10 false alarms of 96, no conformance-only scenario	All 47 scenarios “detected” at 28 false alarms of 96; 6 conformance-only	F-X19.b and F-X19.d
X19 A1	Does the amended fan coil onboarding hold?	Budget restored and no conformance-only detection	41 of 47 detected at 4 false alarms of 96; 0 conformance-only	None
X20	Is model support the log’s own heating-day count across four buildings?	The identity holds on every building; $\rho < 0$ at $p < 0.05$	$\rho = -1$, exact one-sided $p = 1/24 = 0.042$; the identity misses by 44 days on the fan coil unit	F-X20.a and F-X20.b
X21	Can a net’s reading of a foreign year escape being that log’s count?	The bound holds, the tautology persists, the ordering matches the count	The tautology fails on 6 of 12 pairs; the foreign predictor gives $\rho = -0.6$ against the count’s -1	F-X21.a; P3 failed without one
X22	Does the conformance channel fire on injected order violations?	Reversal is significant on the single-duct unit	Significant on 2 of 5 systems; the single-duct hit rate is 0 at zero false alarms	None; P1–P4 failed
X24	Does an outdoor-air-fraction residual add detections?	The 20 % damper leak is detected, budget held, nothing lost	Fan coil 41 to 42 of 47 at 4 false alarms of 96; no system loses a detection	None
X25 steps 1–2	Are the conformance thresholds calibrated out of sample?	Rows out of sample, budget ≤ 10 of 96, at most 3 verdict flips	All held at 0 verdict flips; the in-sample rows had understated false alarms by 3 and 4 days	None
X25 step 3	Does a $\max(\text{fp}, 3)/n$ floor on the pooled denominator keep verdicts?	Counts within 2 and at most 3 verdict flips	Held at 2 verdict flips; one dual-duct scenario gained, one series-unit scenario lost	None
X25 step 4	Does a per-day residual null keep the detections?	Each system’s count stays within 2	The dual-duct unit loses 3 detections (the series unit 2, the parallel and fan coil units 1 and 2) and 8 verdicts flip across the five systems; budgets unchanged (step-level artefact, before the fan-law channel; the final regeneration, x25_statistical_repair.json, nets seven lost and one gained on the dual-duct unit, the count Section 9 quotes, and fires F-X25.c on the series unit)	F-X25.c
X26 (+A1)	Does the false-alarm budget hold on a real building?	At most 3 silence iterations, false alarms under 10 %, the fault case flagged	6.1 % false-alarm days with every channel in budget; the 40 % undercharge case is not flagged	None; both case predictions failed
X27	Does a fan-law virtual static recover static-pressure biases?	At least 7 of 8 static-bias scenarios detected	6 of 8 detected; budget held and nothing lost	F-X27.b
X28	Does a water-side coil UA channel see the simulated fouling?	At least 3 dual-duct and 2 fan coil fouling scenarios gained	0 gained on either system; run 1 also broke the fan coil budget	F-X28.b; run 1 also F-X28.a
X29	Is fouling visible when the refrigerant side is recorded?	At least 3 of 5 condenser and 3 of 5 evaporator severities detected, monotone	All 10 fouling severities detected and monotone at 0 false alarms of 28	None
X30	Does any other process-mining instrument see	No cell detects a miss within 3 false alarms	33 cells against 33 missed scenarios, 0 detections; worst	F-X30.b (two cells not